Switch points of bi-persistence matching distance

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Abstract

In multi-parameter persistence, the matching distance is defined as the supremum of weighted bottleneck distances on the barcodes given by the restriction of persistence modules to lines with a positive slope. In the case of finitely presented bi-persistence modules, all the available methods to compute the matching distance are based on restricting the computation to lines through pairs from a finite set of points in the plane. Some of these points are determined by the filtration data as they are entrance values of critical simplices. However, these critical values alone are not sufficient for the matching distance computation and it is necessary to add so-called *switch points*, i.e. points such that on a line through any of them, the bottleneck matching switches the matched pair.

This paper is devoted to the algorithmic computation of the set of switch points given a set of critical values. We find conditions under which a candidate switch point is erroneous or superfluous. The obtained conditions are turned into algorithms that have been implemented. With this, we analyze how the size of the set of switch points increases as the number of critical values increases, and how it varies depending on the distribution of critical values. Experiments are carried out on various types of bi-persistence modules.

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1 Introduction

The exact computation of the matching distance for multi-parameter persistence modules is an active area of research in computational topology. Besides the theoretical interest [5, 6, 10], one practical reason is that efficient computation of this distance would open the way to the actual application of multi-parameter persistent homology in topological data analysis tasks [3, 8, 11, 14].

The matching distance is defined by taking the supremum of the weighted bottleneck distances on all the (infinitely many) lines with positive slope in the parameter space [7]:

▶ **Definition 1** (Matching Distance). *The* matching distance *between the n-persistence modules M and N is defined by*

$$d_{match}(M,N) := \sup_{L} \hat{m}^L \cdot d_B(\operatorname{dgm} M^L, \operatorname{dgm} N^L)$$

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where $L\colon u = s\vec{m}+b$ is any line, with direction $\vec{m}$ such that $\hat{m}^L := \min_i m_i > 0$, parametrized by $s \in \mathbb{R}$, taken with the standard normalization such that $\max_i m_i = 1$ and $\sum_i b_i = 0$, M^L and N^L are the restrictions of M and N, respectively, to L, dgm *denotes the associated persistence diagram, and d_B the bottleneck distance of persistence diagrams.*

All the available methods for the exact computation of the matching distance are confined to the case $n = 2$, i.e. the matching distance on bi-persistence modules (cf., chronologically, [12, 4, 2]). This will also be the setting of this paper. Another commonality among these methods is that they make use of a finite set of lines through two kinds of points in the plane.

On one hand, [12, 4, 2] use sets of critical grades, such as the sets of entrance grades C_M and C_N of the simplices in the filtration defining the persistence modules M and N respectively, hereafter called *critical values*, and their closure with respect to the least upper bound. These points are directly available from the input data. In [12, 13] critical values are used to define an initital line arrangement in the dual plane, which is then refined.

In our work, critical values are used to define an equivalence relation $\sim_{\overline{C}_M \cup \overline{C}_N}$ on lines by setting $L \sim_{\overline{C}_M \cup \overline{C}_N} L'$ when L, L' partition this set of points in the same way. Interestingly, for any two lines L, L' in the same class, the persistence diagrams dgm M^L and dgm $M^{L'}$ (respectively dgm N^L and dgm $N^{L'}$) are in bijection, as shown, e.g., in [1, Thm. 2]. Moreover, these bijections extend to a bijection, denoted $\Gamma_{L,L'}$, between the set of matchings between M^L and N^L and the set of matchings between $M^{L'}$ and $N^{L'}$, as shown, e.g., in by [2].

Ideally, one would like to use the bijection $\Gamma_{L,L'}$ to show that a line L on which $d_B(M^L, N^L)$ attains a maximum is a line that passes through two points in $\overline{C}_M \cup \overline{C}_N$; unfortunately this is not the case, as shown in [2, Section 2.3]. In fact, it is possible that the matched pair of points that obtains the bottleneck distance may "switch" within a $\sim_{\overline{C}_M \cup \overline{C}_N}$ equivalence class. Therefore, the equivalence relation $\sim_{\overline{C}_M \cup \overline{C}_N}$ must be refined, in order to generate equivalence classes such that there exists at least one matching and matched pair which computes the bottleneck distance for all lines in a given equivalence class. This requires the addition of a second type of points, hereafter called *switch points*: they are points ω in the projective completion such that the cost of matching some pair u, v equals the cost of matching some other pair w, x, with $u, v, w, x \in \overline{C}_M \cup \overline{C}_N$ for any line through ω.

As proven in [2], there exists a finite set of points $\Omega = \Omega(M, N)$ in $\mathbb{P}^2$ with the following property: once the equivalence relation $\sim_{\overline{C}_M \cup \overline{C}_N}$ is refined to $\sim_{\overline{C}_M \cup \overline{C}_N \cup \Omega}$, at least one matching σ and matched pair $u, v \in \overline{C}_M \cup \overline{C}_N$ determines the bottleneck distance for all lines in a given equivalence class. Therefore, the matching distance between M and N may be computed by considering only lines through pairs of points in $\overline{C}_M \cup \overline{C}_N \cup \Omega$.

It is important to underline that the formulas derived in [2] for the points of Ω may produce many unnecessary points. The goal of this paper is to achieve a minimal set of switch points so as to reduce the number of lines needed to compute the matching distance. To this end, we examine the configurations of four points, not necessarily distinct, used to produce candidate switch points. Each such configuration can produce a set of candidate switch points on the order of $10^3 n^4$ where $n = |C_M \cup C_N|$. We identify configurations that cannot produce switch points, as there is no line L with positive slope separating them correctly (cf. Propositions 8 and 12). We also identify candidate points that are superfluous based on geometrical considerations (cf. Propositions 10, 13, and 14). For the proofs of these results, we refer the reader to Appendix A.

Based on these results, we create functions (given in Appendix B) used to discard configurations of points u, v, w, x that cannot give rise to any switch point and candidate switch points that are superfluous. These functions are implemented in a series of algorithms which, together, generate a complete set of switch points.

We have implemented these algorithms and tested them in various experiments, varying the number and distribution of input critical values. Our tests show that, after the geometric pruning of candidate switch points, the number of unique switch points generated by our algorithms is reduced from the theoretical upper bound by a factor of $3.58 \cdot 10^4$.

Organization of the paper. In Section 2 we recall the connection between critical values of bi-persistence modules and points in the persistence diagram of their restriction to a line with positive slope. In Section 3 we review the formulas from [2] to compute candidate switch points. Section 4 contains the geometrical results used to refute some candidate switch points as erroneous or superfluous. Finally, in Section 5 we present the experimental results. All proofs are in Appendix A and pseudocodes are in Appendix B.

2 Preliminaries

We specialize our treatment of persistence to the cases with 1 or 2 parameters, referring to them as persistence or bi-persistence, respectively, throughout the paper.

▶ **Definition 2** (Parameter Spaces). *$\mathbb{R}$ with the usual order $\leq$ is the $1D$-parameter space. The poset $(\mathbb{R}^2, \preceq)$ equipped with the following partial order is called $2D$-parameter space: for $u = (u_1, u_2), v = (v_1, p_2) \in \mathbb{R}^2$, $u \preceq v$ if and only if $u_i \leq v_i$ for $i = 1, 2$.*

The $2D$-parameter space can be sliced by lines with positive slope. A line $L \subset \mathbb{R}^2$ is a $1D$-parameter space when considered parameterized by $s \in \mathbb{R}$ as $L : u = \vec{m}s + b$ where $\vec{m} \in \mathbb{R}^2$ and $b \in \mathbb{R}^2$. L has positive slope if each coordinate m_i of $\vec{m}$ is strictly positive.

▶ **Definition 3** (Persistence Modules). *Let $\mathbb{F}$ be a fixed field and $n \in \{1, 2\}$. An n-*persistence module *M over the parameter space $\mathbb{R}^n$ is an assignment of an $\mathbb{F}$-vector space M_u to each $u \in \mathbb{R}^n$, and linear maps, called* transition *or* internal maps*, $i_M(u, v) \colon M_u \to M_v$ to each pair of points $u \preceq v \in \mathbb{R}^n$, satisfying the following properties:*

- *$i_M(u, u)$ is the identity map for all $u \in \mathbb{R}^n$.*
- *$i_M(v, w) \circ i_M(u, v) = i_M(u, w)$ for all $u \preceq v \preceq w \in \mathbb{R}^n$.*

In this paper, bi-persistence modules will always be assumed to be finitely presented. In particular, for such a module M, there exists a finite set C_M of points in $\mathbb{R}^2$, called *critical values*, for which the ranks of transition maps $i_M(u, v)$ are completely determined by the ranks of transition maps between values in $\overline{C}_M$, the closure of C_M under least upper bound. For example, one can take C_M to be the set of grades of generators and relators of M. Alternatively, C_M can be the set of entrance values of cells in a bi-filtration of a cell complex whose homology gives M.

The *restriction* of M to a line of positive slope $L \subseteq \mathbb{R}^2$ is the persistence module M^L that assigns M_u to each $u \in L$, and whose transition maps $i_{M^L}(u, v) \colon (M^L)_u \to (M^L)_v$ for $u \preceq v \in L$ are the same as in M. Once a parameterization $u = \vec{m}s + b$ of L is fixed, the persistence module M^L is isomorphic to the 1-parameter persistence module, by abuse of notation still denoted by M^L, which

- assigns to each $s \in \mathbb{R}$ the vector space $(M^L)_s := M_u$, and
- whose transition map between $(M^L)_s$ and $(M^L)_t$ for $s \leq t$ is equal to that of M between M_u and M_v with $u = \vec{m}s + b$ and $v = \vec{m}t + b$.

As M^L and N^L are finitely presented persistence modules, they can be uniquely decomposed as a finite sum of interval modules [15]. A module with underlying support the interval $[b, d)$ can be represented as a point (b, d) in $\mathbb{R} \times (\mathbb{R} \cup \{\infty\})$, lying above the diagonal. This encoding permits defining persistence diagrams and the *bottleneck distance* d_B on them [9].

▶ **Definition 4** (Persistence Diagram). *If* $M \cong \oplus_{j\in J} I[b_j, d_j)^{r_j}$, *then the* persistence diagram *of* M, *denoted* $\operatorname{dgm} M$, *is the finite multi-set of points* (b_j, d_j) *of* $\mathbb{R} \times (\mathbb{R} \cup \{\infty\})$ *with multiplicity* r_j *for* $j \in J$.

▶ **Definition 5** (Bottleneck Distance). *Let* M, N *be two* 1*-parameter persistence modules, with persistence diagrams* $\operatorname{dgm} M$ *and* $\operatorname{dgm} N$. *A* matching *between* $\operatorname{dgm} M$ *and* $\operatorname{dgm} N$ *is a multi-bijection* $\sigma : P \to Q$ *between the points of a multi-subset* P *in* $\operatorname{dgm} M$ *and a multi-subset* Q *in* $\operatorname{dgm} N$. *The* cost of a matching σ, *denoted* $c(\sigma)$, *is*

$$c(\sigma) := \max\left\{ \max_{p\in P} \|p - \sigma(p)\|_\infty, \max_{p\notin P \coprod Q} \frac{p_2 - p_1}{2} \right\}.$$

The bottleneck distance d_B *is then*

$$d_B(\operatorname{dgm} M, \operatorname{dgm} N) := \min_{\substack{\sigma: P\to Q \\ P\subseteq \operatorname{dgm} M, Q\subseteq \operatorname{dgm} N}} c(\sigma).$$

Briefly, we can say that $c(\sigma)$ is attained by some $\frac{|s-t|}{\delta}$, with $\delta \in \{1, 2\}$ and s, t coordinates of points in $\operatorname{dgm} M \cup \operatorname{dgm} N$.

When the number of parameters is $n = 2$, we can use the bottleneck distance to define the matching distance between persistence modules M and N as in Definition 1 via restrictions to lines with positive slope. In this case, for convenience, we set $\operatorname{cost}(\sigma^L) := \hat{m}^L c(\sigma^L)$ where σ^L is a matching between $\operatorname{dgm} M^L$ and $\operatorname{dgm} N^L$ and $\hat{m}^L = \min_i m_i$.

As shown in [1], it is possible to partition the lines of $\mathbb{R}^2$ with positive slope into equivalence classes with respect to the relation $\sim_{\overline{C}_M \cup \overline{C}_N}$ so that, for any two lines L, L' in the same class, the persistence diagrams $\operatorname{dgm} M^L$ and $\operatorname{dgm} M^{L'}$ (respectively $\operatorname{dgm} N^L$ and $\operatorname{dgm} N^{L'}$) are in bijection. Moreover, by [2], these bijections extend to a bijection, denoted $\Gamma_{L,L'}$, between the set of matchings σ^L between M^L and N^L and the set of matchings $\sigma^{L'}$ between $M^{L'}$ and $N^{L'}$. We now review this equivalence relation defined in [1].

▶ **Definition 6** (Positive Cone). *For every point* u *in* $\mathbb{R}^2$, *let* $S_+(u)$ *be the* positive cone with vertex u: $S_+(u) := \{v \in \mathbb{R}^2 : u \preceq v\}$. *The boundary of the positive cone,* $\partial S_+(u)$, *decomposes into open faces: For* $\emptyset \neq A \subseteq \{1, 2\}$, *define*

$$S_A(u) := \begin{cases} \{u\} & \text{if } A = \{1, 2\}\ , \\ \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 = u_1, x_2 > u_2\} & \text{if } A = \{1\}\ , \\ \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 > u_1, x_2 = u_2\} & \text{if } A = \{2\}\ . \end{cases}$$

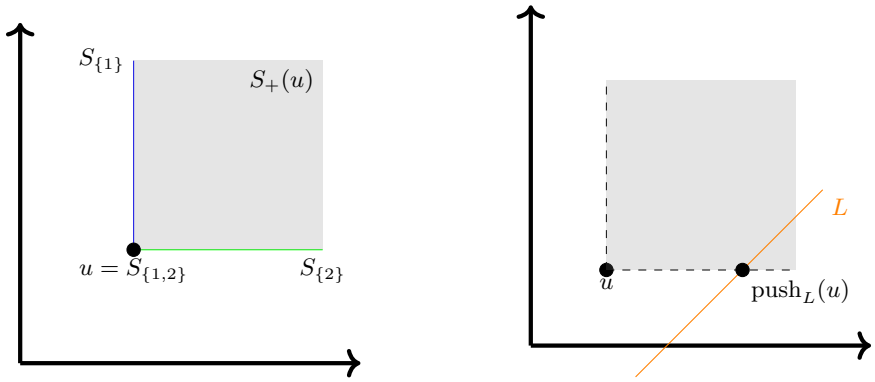


Figure 1 Left: The positive cone $S_+(u)$ of $u \in \mathbb{R}^2$ and the decomposition of its boundary into $S_{\{1\}}$, $S_{\{2\}}$ and $S_{\{1,2\}}$. Right: The push of u along the line L when $A_u^L = \{2\}$ (u to the left of L).

▶ **Definition 7.** *A line L with positive slope intersects $\partial S_+(u)$ in exactly one point, which we call the* push of u onto L, *denoted* $\text{push}_L(u) := L \cap \partial S_+(u)$. *We use the notation $p_L(u)$ for the parameter value of the push of a point u onto the line L.*

Because of the partition of $\partial S_+(u)$ by its open faces, there is a unique non-empty subset of $\{1, 2\}$, denoted A_u^L, such that $\text{push}_L(u) = L \cap S_{A_u^L}(u)$. Concretely, in the plane, $A_u^L = \{1\}$ means that u is strictly to the right of L, $A_u^L = \{2\}$ means that u is strictly to the left of L, $A_u^L = \{1, 2\}$ means that u is on L. More generally, when $\{1\} \subseteq A_u^L$, we say u is to the right of L, and when $\{2\} \subseteq A_u^L$, we say u is to the right of L, allowing for the fact that it may be true in either of these cases that u is on L. We say that two lines $L, L' \subseteq \mathbb{R}^2$ with positive slope have the *same reciprocal position* with respect to u if and only if $A_u^L = A_u^{L'}$. Given a non-empty subset U of $\mathbb{R}^2$, we write $L \sim_U L'$ if L and L' have the same reciprocal position with respect to u for all $u \in U$. Note that $\sim_U$ is an equivalence relation on lines with positive slope in $\mathbb{R}^2$.

As shown in [2], we need to extend this equivalence relation to include points in the projective completion of the real plane, $\mathbb{P}^2 = \mathbb{R}^2 \cup \ell_\infty$, with ℓ_∞ the line at infinity of $\mathbb{P}^2$. Points on the line at infinity are given by homogeneous coordinates $[x_0 : x_1 : x_2]$ with $x_0 = 0$. Note that a line with positive slope in $\mathbb{R}^2$ will intersect ℓ_∞ at exactly one point $[0 : v_1 : v_2]$ with $\vec{v} = (v_1, v_2) \succ 0$ giving the direction of the line. Given a point $v = [0 : v_1 : v_2] \in \ell_\infty$ with $v_1, v_2 > 0$, define

$$S_A(v) := \begin{cases} \{v\} & \text{if } A = \{1, 2\} \ , \\ \{[0 : x_1 : x_2] \in \ell_\infty \mid x_1 = v_1, x_2 > v_2\} & \text{if } A = \{1\} \ , \\ \{[0 : x_1 : x_2] \in \ell_\infty \mid x_1 > v_1, x_2 = v_2\} & \text{if } A = \{2\} \ . \end{cases}$$

For every such $v \in \mathbb{P}^2$ and line $L \subset \mathbb{R}^2$ with positive slope, define A_v^L to be the largest subset of $\{1, 2\}$ such that $L \cap S_{A_v^L} \neq \emptyset$. As above, we say that two lines L, L' in $\mathbb{R}^2$ with positive slope are in the *same reciprocal position* with respect to v if and only if $A_v^L = A_v^{L'}$.

We then extend the equivalence relation $\sim_U$ form $\mathbb{R}^2 \times \mathbb{R}^2$ to $\mathbb{P}^2 \times \mathbb{P}^2$ as follows: two lines L and L' are equivalent if and only if they are in the same reciprocal position with respect to a finite set U of points in $\mathbb{P}^2$: $L \sim_U L'$ if and only if $A_u^L = A_u^{L'}$ for all $u \in U$.

3 Explicit formulas for switch points

Let C_M and C_N denote the set of critical values of M and N, respectively, as guaranteed by the assumption that M and N are finitely generated, and let $\overline{C}_M$ and $\overline{C}_N$ be their closure with respect to the least upper bound.

First, we note that all candidate switch points ω, either proper or at infinity, can be computed by checking quadruples u, v, w, x (of which at least three must be distinct) $C_M \cup C_N$ assuming that u, v and w, x are matched pairs for which, calling U this set of points,

- there is a line L in a particular equivalence class E via $\sim_U$ on which the cost of matching u with v equals the cost of matching w with x, but
- the cost of matching u with v *does not* equal the cost of matching w with x for all lines in that equivalence class.

All such lines with this equality in E pass through some point $\omega \in \mathbb{P}^2$, as shown in Figures 2 and 3. Spanning over all quadruples and all equivalence classes via $\sim_{\overline{C}_M \cup \overline{C}_N}$, we obtain the set of candidate switch points $\Omega(M, N)$. To avoid degenerate or trivial cases, we assume that $u \neq v$ and $w \neq x$. This is because the matched pair u, v with $u = v$ would indicate a simultaneous birth and death, i.e. a point on the diagonal within a persistence diagram that yields a null contribution to the matching distance (similarly for $w = x$). However, it can be the case that one of the four points is repeated.

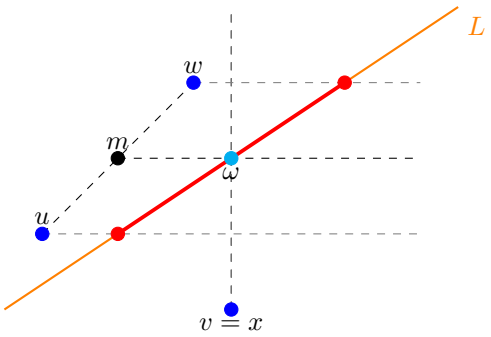


Figure 2 An example of 3 points $u, w, v = x$ that generate an ω point with m denoting the midpoint of v and x.

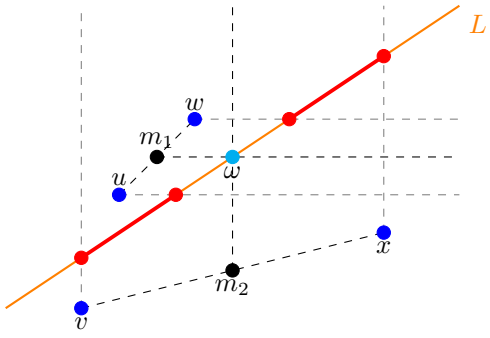


Figure 3 An example of 4 points u, w, v, x which generate an ω point with m_1 and m_2 denoting the midpoints of u and w, and of v and x, respectively.

We also use the following notation: For U a non-empty set of points in the plane, we set $A(U) = \{A_u \mid u \in U, \emptyset \neq A_u \subseteq \{1, 2\}\}$. The set $A(U)$ specifies, for each $u \in U$, one or more push directions, and there are 3^m possible choices for $A(U)$ if $|U| = m$. Given such a set $A(U)$, we define $E(A(U))$ to be the set of all lines L with positive slope such that $A_u^L = A_u$ for each $u \in U$, i.e. all points of U push onto these lines in the directions specified by $A(U)$. Note that it is a union of all equivalence classes of $\sim_{\overline{C}_M \cup \overline{C}_N}$ in which the points of U push in the specified directions.

We introduce this notation because switch points are generated based on how the pairs of points u, v and w, x push onto a line. Taking $U = \{u, v, w, x\}$, this is only a subset of the set $\overline{C}_M \cup \overline{C}_N$, implying that there may be many equivalence classes via $\sim_{\overline{C}_M \cup \overline{C}_N}$ for which these four points push in the same way for each equivalence class. So, we create the union of all such equivalence classes via the set $E(A(u, v, w, x))$.

In [2, Lemma 4.1], we proved that, for fixed $U = \{u, v, w, x\} \subseteq \overline{C}_M \cup \overline{C}_N$ with $u \neq v$ and $w \neq x$, with at least three distinct points among them, fixed $\delta, \eta \in \{1, 2\}$, and fixed a choice for $A_u^L, A_v^L, A_x^L, A_w^L$, if ω is a switch point for $\sim_U$, then it is determined by one of the following formulas: Assuming $p_L(u) \geq p_L(v)$ and $p_L(w) \geq p_L(x)$,

- if u, v, w lie to the left of L and x lies strictly to the right, then

$$\omega = [\delta : \delta x_1 : \eta(v_2 - u_2) + \delta w_2]\,. \tag{1}$$

- if u, v, and w lie to the right of lines $L \in E$, and x to the left, then

$$\omega = [\delta : \eta(v_1 - u_1) + \delta w_1 : \delta x_2]\,. \tag{2}$$

- If u, v lie strictly to the left of L, x, w strictly to the right, $x_1 \neq w_1$, and $u_2 \neq v_2$, then

$$\omega = [0 : \delta(w_1 - x_1) : \eta(u_2 - v_2)]\,. \tag{3}$$

- If u, w lie strictly to the left of L, x, v strictly to the right, $\delta = \eta$, $x_1 \neq v_1$, $u_2 \neq w_2$, then

$$\omega = [0 : v_1 - x_1 : u_2 - w_2]\,. \tag{4}$$

- If u, w lie strictly to the left of L, v, x lie strictly to the right, and $\delta \neq \eta$, then

$$\omega = [\eta - \delta : \eta v_1 - \delta x_1 : \eta u_2 - \delta w_2] \tag{5}$$

Assuming $p_L(u) \geq p_L(v)$ and $p_L(w) \leq p_L(x)$,

- if u, v, w lie to the left of L and x lies strictly to the right, then

$$\omega = [\delta : \delta x_1 : \eta(u_2 - v_2) + \delta w_2]\,. \tag{6}$$

- if u, v, and w lie to the right of lines $L \in E$, and x to the left, then

$$\omega = [\delta : \eta(u_1 - v_1) + \delta w_1 : \delta x_2]\,. \tag{7}$$

- If u, w lie strictly to the left of L, v, x lie strictly to the right:

$$\omega = [\eta + \delta : \delta x_1 + \eta v_1 : \delta w_2 + \eta u_2]\,. \tag{8}$$

These formulas represent a necessary condition for ω to be a switch point in each case. More precisely, they are obtained by imposing that

$$\Delta_L(u, v, w, x; \delta, \eta) := \frac{|p_L(u) - p_L(v)|}{\delta} - \frac{|p_L(w) - p_L(x)|}{\eta}$$

is zero for a line with positive slope L, under the chosen conditions for δ, η, and $A_u^L, A_v^L, A_x^L, A_w^L$.

4 Computation of switch points

To compute all possible switch points, we create algorithms for each of the three cases (listed as cases (2)-(4) in the proof of [2, Lemma 4.1]) for which switch points may be generated. We do not include an algorithm for case (1), as this situation does not produce switch points. Algorithm 3 corresponds to case (2), when three points push in one direction and the fourth point pushes in the other direction; Algorithm 5 corresponds to case (3), when two paired points push in one direction while the other two paired points push in the other direction;

finally, Algorithm 7 corresponds to case (4), when two unpaired points push in one direction while the other two unpaired points push in the other direction.

All algorithms share the common assumptions that, among the four points u, v, w, x u is paired to v and w is paired to x, and at least 3 of the points u, v, w, x are distinct. The values of both δ, η depend on which of C_M or C_N the points u, v, w, x belong to. Each point can belong to either of the sets, yielding for each set of u, v, w, x multiple combinations of δ and η. If u, v are both in C_M (respectively C_N), then we need to set $\delta = 2$, and if they are in different sets, then we need to set $\delta = 1$. The same reasoning applies for w, x and η, yielding up to four possible combinations of $\delta, \eta \in \{1, 2\}$. In the algorithms, this is achieved by considering the module a point comes from as a parent of the point. It is possible that $a \in C_M \cap C_N$. In this case, a admits as parents both M and N.

Some configurations of points u, v, w, x are not feasible and should be discarded. In this paper, we address two situations in which computations would lead to erroneous switch points:

1. there is no line with positive slope that separates the four points as required;
2. the condition $\Delta_L = 0$ cannot be satisfied by any line with positive slope that separates the four points as required;

In what follows, we will make sure that the obtained switch points do not present issues as in Items 1 and 2. We now analyze the three algorithms separately.

4.1 Algorithm 3vs1

Algorithm 3 is used to compute switch points for all possible combinations of four points in $C_M \sqcup C_N$, the corresponding possible combinations of δ and η, and the possible choices of $A(u, v, w, x)$ leading to configurations where the point x is strictly on the left (resp. right) of a line in $E(A(u, v, w, x))$ and the points u, v, w on the right (resp. left) of that line, but not strictly. This corresponds to the case listed as case (2) in the proof of [2, Lemma 4.1].

To address issue 1), we introduce the function CheckPts (see Algorithm 1) as a way of discarding some impossible switch points. The correctness of the function CheckPts is guaranteed by the following proposition which identifies when there are no lines in the equivalence class $E(A(u, v, w, x))$: this occurs when x is contained in the convex hull of u, v, w, or when x is required to push upward (resp. rightward), but it is to the left and above (resp. to the right and below) of one of the three points. The following proposition is used to show that the function CheckPts will correctly determine quadruples for which there is no line that separates the four points as required.

▶ **Proposition 8.** *Given four not necessarily distinct points u, v, w, x, let A_u, A_v, A_w, A_x be non-empty subsets of $\{1, 2\}$. The following statements hold:*

1. *If $2 \in A_u \cap A_v \cap A_w$ and $A_x = \{1\}$, then $E(A(u, v, w, x))$ is empty if and only if there exists $a = (a_1, a_2)$ in the convex hull of u, v, w such that $x_1 \leq a_1$ and $x_2 \geq a_2$.*
2. *If $1 \in A_u \cap A_v \cap A_w$ and $A_x = \{2\}$, then $E(A(u, v, w, x))$ is empty if and only if there exists $a = (a_1, a_2)$ in the convex hull of u, v, w, such that $x_1 \geq a_1$ and $x_2 \leq a_2$.*
3. *If $2 \in A_u \cap A_v \cap A_w$ and $A_x = \{1, 2\}$, then $E(A(u, v, w, x))$ is empty if and only if there exists $a = (a_1, a_2)$ in the convex hull of u, v, w such that $x_1 \leq a_1$, $x_2 \geq a_2$, and $x \neq a$.*
4. *If $1 \in A_u \cap A_v \cap A_w$ and $A_x = \{1, 2\}$, $E(A(u, v, w, x))$ is empty if and only if there exists $a = (a_1, a_2)$ in the convex hull of u, v, w such that $x_1 \geq a_1$, $x_2 \leq a_2$, and $x \neq a$.*

To address issue 2), we introduce the function CheckOmega (see Algorithm 2) as a way of excluding the switch points which are generated by a quadruple that can be separated appropriately by a line with positive slope, but for which no such line satisfies $\Delta_L = 0$. Proposition 9 determines these quadruples; Proposition 10 shows Algorithm 1 may be used to partially check the conditions of Proposition 9.

▶ **Proposition 9.** *Given three not necessarily distinct points u, v, w, and a point $x \neq w, u, v$, let A_u, A_v, A_w, A_x be non-empty subsets of $\{1, 2\}$ for which $E(A(u, v, w, x))$ is non-empty.*

1. *When $2 \in A_u \cap A_v \cap A_w$ and $A_x = \{1\}$, letting ω be given by Equation (1) or Equation (6), $E(A(u, v, w, x))$ contains some lines L for which $\Delta_L(u, v, w, x; \delta, \eta) = 0$ if and only if*
   - *$\omega_2 > x_2$ and, moreover,*
   - *there is no $a = (a_1, a_2)$ in the convex hull of u, v, w with $\omega_1 \leq a_1$, $\omega_2 \geq a_2$, and $a \neq \omega$.*
2. *When $1 \in A_u \cap A_v \cap A_w$ and $A_x = \{2\}$, letting ω be given by Equation (2) or Equation (7), $E(A(u, v, w, x))$ contains some lines L for which $\Delta_L(u, v, w, x; \delta, \eta) = 0$ if and only if*
   - *$\omega_1 > x_1$ and, moreover,*
   - *there is no $a = (a_1, a_2)$ in the convex hull of u, v, w with $\omega_1 \geq a_1$, $\omega_2 \leq a_2$, and $a \neq \omega$.*

▶ **Proposition 10.** *Given three not necessarily distinct points u, v, w, the following statements are equivalent:*

- *there is no $a = (a_1, a_2)$ in the convex hull of u, v, w with $\omega_1 \leq a_1$, $\omega_2 \geq a_2$, and $a \neq \omega$.*
- $\text{CheckPts}(u, v, w, \omega, \text{Below}) = \text{True}$, *or* $\text{CheckPts}(u, v, w, \omega, \text{Below}) = \text{False}$ *and* $\omega \in \partial\text{convhull}(u, v, w)$.

Similarly, the following statements are equivalent:

- *there is no $a = (a_1, a_2)$ in the convex hull of u, v, w with $\omega_1 \geq a_1$, $\omega_2 \leq a_2$, and $a \neq \omega$.*
- $\text{CheckPts}(u, v, w, \omega, \text{Above}) = \text{True}$, *or* $\text{CheckPts}(u, v, w, \omega, \text{Above}) = \text{False}$ *and* $\omega \in \partial\text{convhull}(u, v, w)$.

We observe that if ω is on the boundary of the convex hull of u, v, w, then the line through ω separating u, v, w from x for which $\Delta_L = 0$ passes through two of the points u, v, w. Thus, this line is eventually generated even if ω is not added to the set of switch points as it is a line through two critical points. So, this omega point is superfluous and does not need to be added to the set of switch points. Propositions 9 and 10, and the last observation justify the call of Algorithm 2 by Algorithm 3 to reject erroneous or superfluous switch points.

Finally, in Algorithm 3, given a configuration of points c_1, c_2, c_3, c_4 extracted from $C_M \sqcup C_N$, such that 3 are distinct, we label them as x, u, v, w so that x and w are distinct, and can then relabel (if necessary) the remaining two points to ensure that $p_L(u) > p_L(v)$. To determine which switch point is generated, we must also determine the sign of $p_L(x) - p_L(w)$. The following proposition can be used to determine, from x and w, the possible signs of this difference, ensuring the correctness of lines 28–43 of Algorithm 3.

▶ **Proposition 11.** *Let x, w be two distinct points in $C_M \cup C_N$. For every line L with positive slope which separates x and w, it holds that:*

- *If* l.u.b.$(w, x) = x$, *then* $p_L(x) \geq p_L(w)$.
- *If* l.u.b.$(w, x) = w$, *then* $p_L(x) \leq p_L(w)$.
- *If* l.u.b.(w, x) *is neither x nor w, then $p_L(x) \geq p_L(w)$ for all lines L which separate w and* l.u.b.(w, x), *and $p_{L'}(x) \leq p_{L'}(w)$ for all lines L' which separate x and* l.u.b.(w, x).

4.2 Algorithm 2paired-vs-2paired

Algorithm 5 is used to compute switch points for all possible combinations of four points in $C_M \sqcup C_N$, the corresponding possible combinations of δ and η, and the possible choices of $A(u, v, w, x)$ leading to configurations where two paired points lie strictly on the left of a line in $E(A(u, v, w, x))$ and the other two paired points lie strictly on the right of that line. This corresponds to the case listed as case (3) in the proof of [2, Lemma 4.1].

In case (3) of [2, Lemma 4.1], we saw that if $A_u = A_v = \{2\}$ and $A_x = A_w = \{1\}$, then no switch point is generated if $x_1 = w_1$ or $u_2 = v_2$. Moreover, since the pairs must lie strictly on one side of the line, it is not possible that one point may simultaneously be in two pairs. Since paired points must be distinct (otherwise, the cost of matching that pair would be zero), it must be the case that all 4 points are distinct.

If a quadruple of distinct points generates a switch point ω with this configuration, it is given by Equation (3). Note that, given four distinct points that are separated by a line L with positive slope so that two lie strictly on one side of the line and the other two lie strictly on the other side, there is exactly one way to label these points as x, w, u, v so that x, w push up to L and $p_L(w) > p_L(x)$, and u, v push right to L and $p_L(u) > p_L(v)$. With this labeling, the slope that is generated by Equation (3) is by construction positive.

Next, we introduce the function CheckPts2 (see Algorithm 4) to discard sets of points which cannot be separated by a line with positive slope as required (issue 1). The correctness of this function is guaranteed by Proposition 12. More precisely, the function CheckPts2 uses condition 4 from Proposition 12 to check for erroneous quadruples of points.

▶ **Proposition 12.** *Let u, v, w, x be four points, with $A_u = A_v = \{2\}$, $A_x = A_w = \{1\}$. Set*

$$Q = \{(p_1, p_2) \in \mathbb{R}^2 | p_1 \leq a_1 \text{ and } p_2 \geq a_2 \text{ for some } a \in \overline{uv}\} \text{ and}$$

$$R = \{(p_1, p_2) \in \mathbb{R}^2 | p_1 \geq b_1 \text{ and } p_2 \leq b_2 \text{ for some } b \in \overline{xw}\}.$$

The following statements are equivalent:

(i) $R \cap Q \neq \emptyset$,
(ii) *the set of lines $E(A(u, v, w, x))$ is empty,*
(iii) *there are a point $a = (a_1, a_2)$ on the line segment $\overline{uv}$ and a point $b = (b_1, b_2)$ on the line segment $\overline{xw}$ such that $(b_1 \leq a_1) \wedge (b_2 \geq a_2)$,*
(iv) *Either x or $w \in Q$, or u or $v \in R$.*

The following proposition determines if it is possible to split four points appropriately with a line with slope determined by the ω given in Equation (3) (issue 2). This will happen exactly when the y-intercepts of the lines through u, v, w, x are ordered appropriately. Proposition 13 ensures the correctness of line 31 in Algorithm 5.

▶ **Proposition 13.** *Given four distinct points u, v, w, x, let $A_u = A_v = \{2\}$ and $A_w = A_x = \{1\}$, and assume that the set $E(A(u, v, w, x))$ is non-empty. Without loss of generality, assume that $p_L(u) > p_L(v)$ and $p_L(w) > p_L(x)$. Then, for every $\eta, \delta \in \{1, 2\}$, letting $m = \frac{\delta(w_1 - x_1)}{\eta(u_2 - v_2)}$ be the slope identified by ω as given in Equation (3), the equivalence class $E(A(u, v, w, x))$ contains some lines L for which $\Delta_L(u, v, w, x; \delta, \eta) = 0$ if and only if*

$$\max\{x_2 - mx_1, w_2 - mw_1\} < \min\{u_2 - mu_1, v_2 - mv_1\} \tag{9}$$

4.3 Algorithm 2unpaired-vs-2unpaired

Algorithm 7 is used to compute switch points for all possible combinations of four points in $C_M \sqcup C_N$, with at least three of them distinct, the corresponding possible combinations of δ and η, and the possible choices of $A(u,v,w,x)$ leading to configurations where a line in $E(A(u,v,w,x))$ splits both pairs; i.e., one point of a pair is strictly on the left, and one point is strictly on the right. This corresponds to the case listed as case (4) in the proof of [2, Lemma 4.1]. We may assume without loss of generality that u and w lie to the left of the line, and that v and x lie to its right. Depending on L, both $p_L(u) \geq p_L(v)$ and $p_L(u) \leq p_L(v)$ can occur (similarly for x, w). To decide which one occurs, we apply Proposition 11 in lines 28 and 42 of Algorithm 7.

To address issue 1) we use again the function CheckPts2 (Algorithm 4) simply by changing the roles of u,v,w,x; calling CheckPts2(x,v,w,u) instead of CheckPts2(x,w,u,v). If the configuration is feasible, we compute the candidate switch points with the following formulas:

- If $p_L(u) \geq p_L(v)$ and $p_L(w) \geq p_L(x)$, or $p_L(u) \leq p_L(v)$ and $p_L(w) \leq p_L(x)$, and moreover
  - $\delta = \eta$, and $x_1 = v_1$ or $u_2 = w_2$, then no switch point is generated;
  - $\delta = \eta$, $x_1 \neq v_1$, and $u_2 \neq w_2$, then ω is given by Equation (4);
  - $\delta \neq \eta$, then ω is given by Equation (5).
- If $p_L(u) \geq p_L(v)$ and $p_L(w) \leq p_L(x)$, or $p_L(u) \leq p_L(v)$ and $p_L(w) \geq p_L(x)$, then ω is given by Equation (8).

In the case when ω is a point at infinity, issue 2) can be addressed as in Proposition 13 with the roles of w and v switched. The only assumption from Proposition 13 that is not guaranteed is $p_L(u) > p_L(v)$ and $p_L(w) > p_L(x)$, which, in the proof, ensures that any line passing through ω has positive slope. So, in Algorithm 7, we discard any ω points which generate lines with non-positive slope, and then check a modified version of Equation (9) (line 32). When ω is a proper point, we address issue 2) by introducing the function CheckOmega2 (see Algorithm 6). The correctness of Algorithm 6 is given by Proposition 14.

▶ **Proposition 14.** *Given four points u,v,w,x, let $A_u = A_w = \{2\}$ and $A_v = A_x = \{1\}$ and assume that $E(A(u,v,w,x))$ is non-empty. Let ω be the point generated by x,v,u,w by Equation (5) or Equation (8). Partition $\mathbb{R}^2$ into quadrants with respect to ω in the following way:*

$$Q_1 = \{(p_1,p_2) \in \mathbb{R}^2 | p_1 > \omega_1 \text{ and } p_2 > \omega_2\}, \quad Q_2 = \{(p_1,p_2) \in \mathbb{R}^2 | p_1 \leq \omega_1 \text{ and } p_2 \geq \omega_2\}$$
$$Q_3 = \{(p_1,p_2) \in \mathbb{R}^2 | p_1 < \omega_1 \text{ and } p_2 < \omega_2\}, \quad Q_4 = \{(p_1,p_2) \in \mathbb{R}^2 | p_1 \geq \omega_1 \text{ and } p_2 \leq \omega_2\}$$

1. *If either x or $v \in Q_2$, or u or $w \in Q_4$, there is no line through ω in $E(A(u,v,w,x))$.*
2. *If $x, v \in Q_4$, $u, w \in Q_2$, and $\omega \notin \{x,v,u,w\}$, all lines through ω with positive slope are in $E(A(u,v,w,x))$.*
3. *Suppose that $x, v \notin Q_2$, $u, w \notin Q_4$, and $\{u,v,x,w\} \cap (Q_1 \cup Q_3) \neq \emptyset$. For $c \in \{u,v,x,w\} \cap (Q_1 \cup Q_3)$, let $0 \leq m_c \leq \infty$ be the slope of the line through ω and c, and define*

$$\text{sign}_1(c) = \begin{cases} +1 & \text{if } c \in Q_1 \\ -1 & \text{if } c \in Q_3 \end{cases} \quad \text{and} \quad \text{sign}_2(c) = \begin{cases} +1 & \text{if } c \in \{x,v\} \\ -1 & \text{if } c \in \{u,w\}. \end{cases}$$

Then there is a line with positive slope through ω in $E(A(u,v,w,x))$ if and only if there exists $0 < m < \infty$ such that, for every $c \in \{u,v,x,w\} \cap (Q_1 \cup Q_3)$,

$$\text{sign}_1(c)\text{sign}_2(c)m_c < \text{sign}_1(c)\text{sign}_2(c)m.$$

5 Experiments

First, Figure 4 shows the outputs of Algorithms 3, 5, and 7 for an example pair of modules.

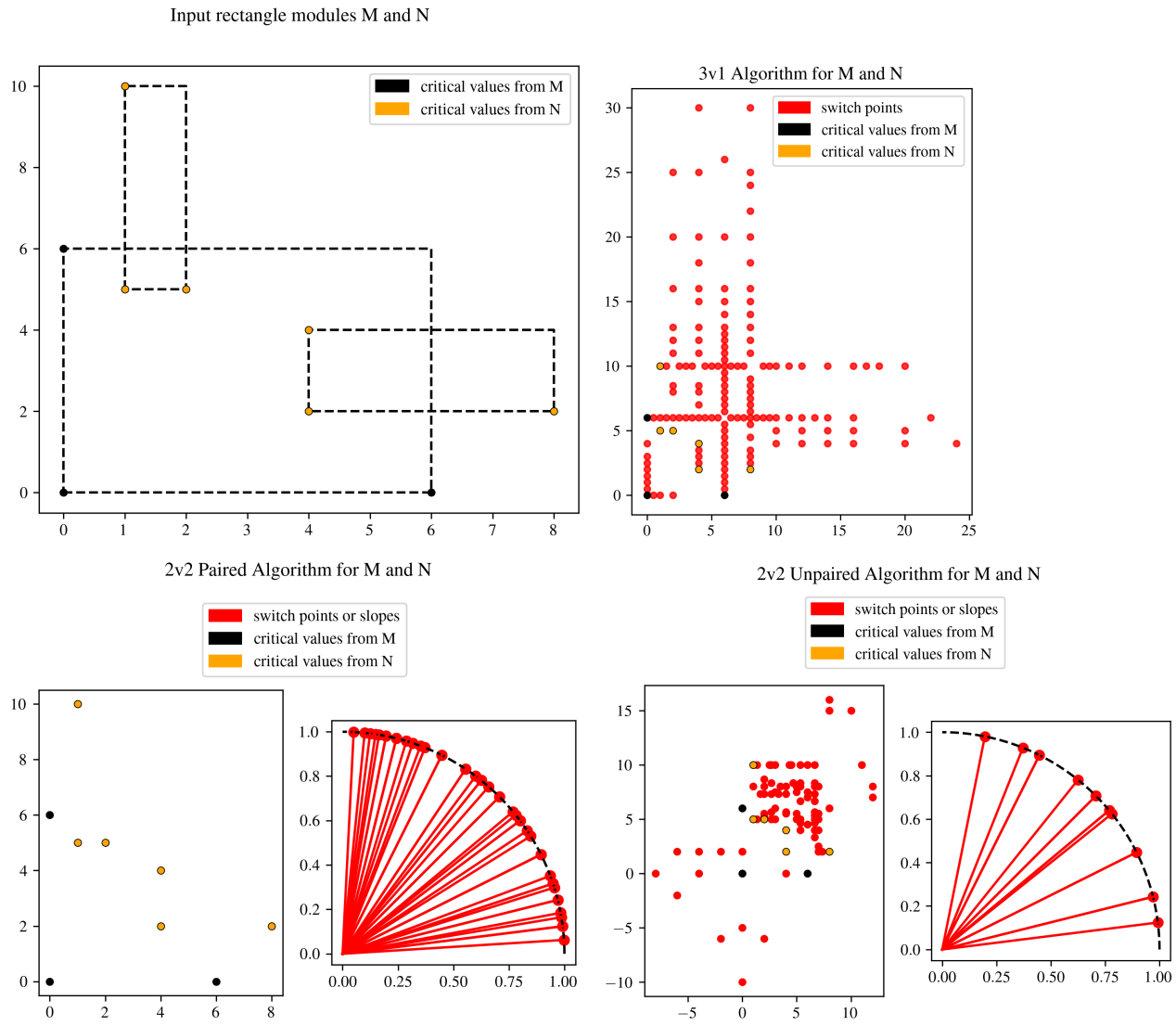


Figure 4 For modules M and N given in the top left, Algorithm 3 computed 914 switch points, and deduplicated them to 142. Algorithm 5 computed 133 slopes (with duplicates), and 44 without duplicates. Note that all of the outputs are points at infinity. Algorithm 7 computed 181 switch points and points at infinity combined (with duplicates) and without duplicates 95.

Next, we consider the theoretical upper bound of candidate switch points. Given two sets of critical values C_M and C_N with $n = |C_M \cup C_N|$, there are $n(n-1)(n-2)(n-3)$ possible subsets of four distinct points c_1, c_2, c_3, c_3, and $n^2(n-1)(n-2)$ possible subsets of four points with one repetition, summing up to $n(n-1)(n-2)(2n-3)$ choices. For each such choice, there are 4 choices of labeling one of them as x or u, and 3 as w or v, yielding 144 choices for each set c_1, c_2, c_3, c_3. As Algorithm 3 produces at most 4 switch points, Algorithm 5 at most 1, and Algorithm 7 at most 2, we deduce that in the worst case the number of switch

points is on the order of $10^3 n(n-1)(n-2)(2n-3)$.

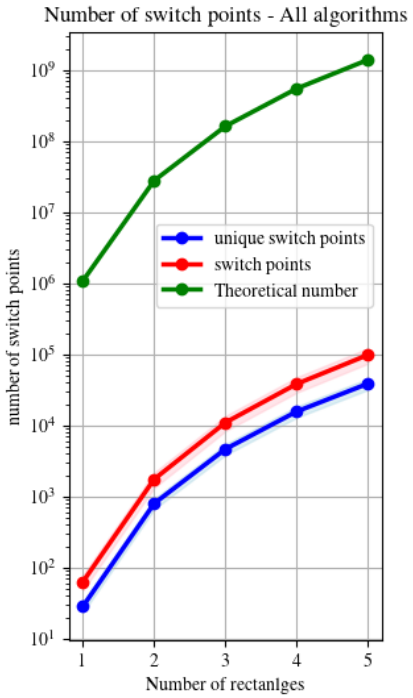


Figure 5 We compute the number of switch points generated for pairs of random rectangle decomposable bi-persistence modules with the same number of rectangles (x-axis). We record the number of switch points, unique switch points and the theoretical upper bound. The average and standard error of the mean are computed over 5 runs.

In Figure 5, we show how the computed number of switch points increases with the number of critical values. We see that the number of unique switch points is smaller than the total number of computed switch points; on average, the ratio of unique points to computed points is 0.55. However, the number of computed switch points is much smaller than the theoretical bound; on average, the ratio of total computed points to the theoretical bound is $6.57 \cdot 10^{-5}$. This is because the Algorithms 3, 5, and 7 discard repeated quadruples which only differ by relabeling, and due to functions CheckPoints, CheckPoints2, CheckOmega, and CheckOmega2.

In Figure 6 we compare the outputs for three data sets: the output of a Poisson point process with blue noise, data with real-world attributes (noisy, less regular), and data from a rectangle module. The results show that on data with fewer symmetry constraints than in the rectangle case, the number of duplicate switch points almost vanishes.

6 Conclusions

Following the theory of [2], we have provided algorithms to generate a set of candidate switch points necessary to compute the matching distance between two finitely generated bi-persistence modules. Further, we have provided conditions and algorithms for excluding erroneous and superfluous candidate switch points, thereby improving the computation of the matching distance (via computing the bottleneck distance of restrictions to lines through

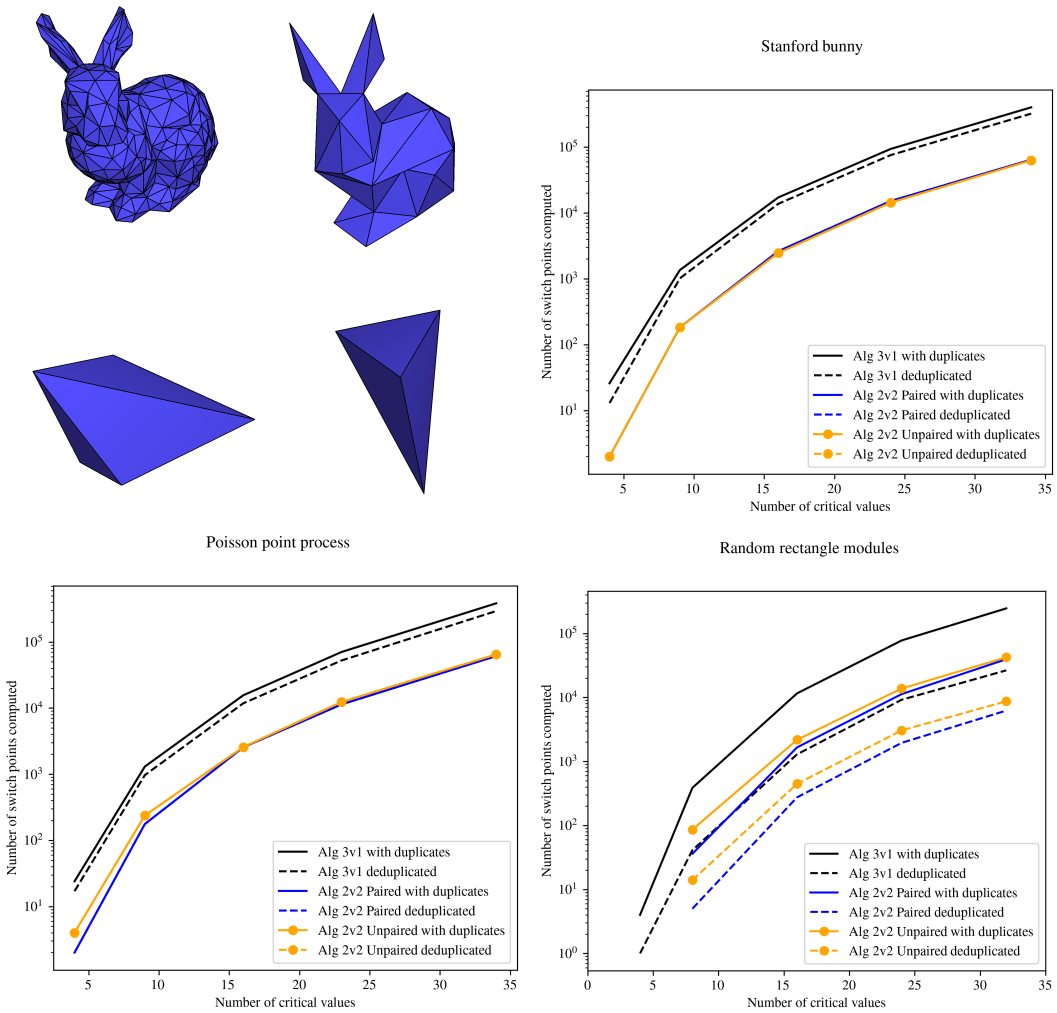


Figure 6 Here we compare the number of switch points generated from differently organised input critical values: namely, the vertices of the decimated Stanford Bunny projected to the xy-plane, the output of a Poisson point process with blue noise, and randomly generated rectangle decomposable bi-persistence modules.

pairs from the union of the closure of critical points and switch points). To our knowledge, while switch points are used in previous work on the computation of the matching distance, the issue of making explicit a minimal set of switch points had not been addressed before.

We have also presented experiments on different types of data which show that our implementation can significantly lower the number of computed switch points from the theoretical bound, showing a slightly different behavior with respect to the number of duplicates.

As any algorithm for the exact computation of the matching distance runs in polynomial time on the number of points used to generate the lines for the restriction, this work has the potential to greatly reduce the overall computation time.

Our results fill an important gap in the computation of the matching distance. However, some questions remain to be answered. First, while we have implemented code to generate the necessary switch points, we have yet to integrate this into a larger algorithm that directly

computes the matching distance between two modules.

Second, while we have been able to discard erroneous and superfluous candidate switch points caused by issues 1 and 2, there are still candidate switch points in the set generated by Algorithms 3, 5, and 7 which can be discarded, because of the following issue. When a candidate switch point ω generated by x, u, v, w lies outside the support of the indecomposable summands associated with x, u, v, w, it may be the case that obtained bars (generated by x, u, v, w) on a line through ω do not actually "switch" matching. The switch point is generated because the cost of matching the projections of the critical values onto the line is equal, but since ω is outside the support, the cost of matching the bars may not be equal.

For example, in Figure 4 (Algorithm 3), there are many switch points outside the support of the critical values with x-coordinate 6. However, the bar that will be generated by the critical values from M will not extend past the support of that rectangle. So, it is likely that such candidate switch points may also be discarded. However, it is not possible to discard such switch points with the assumptions we are making, as this would require knowledge on the support of the persistence modules M and N and their decomposition.

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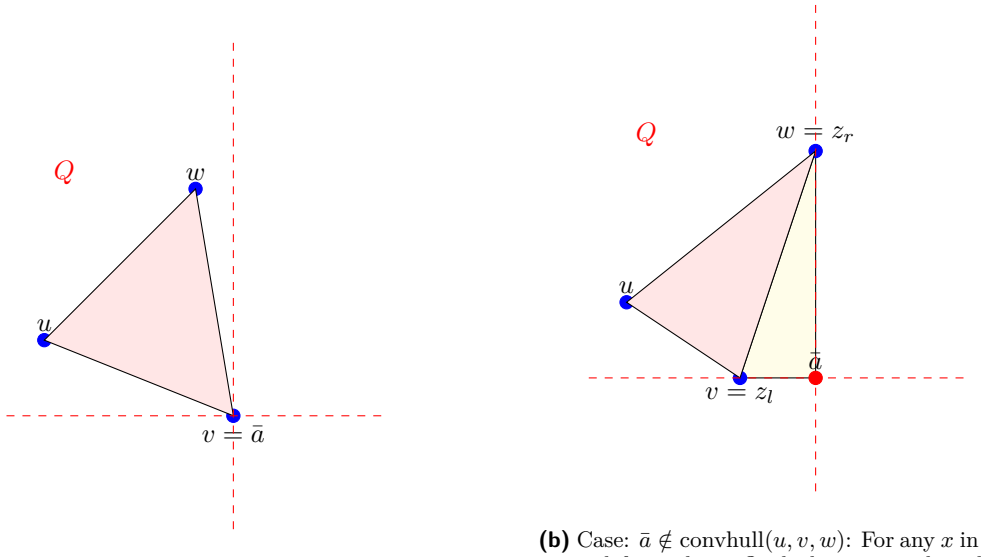
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A Proofs

Proof of Proposition 8. Let us start by proving the first statement. The second one can be proved similarly. We refer to Figure 7 for an illustration.



(a) Case $\bar{a} \in \text{convhull}(u, v, w)$: For any x in the upper left quadrant Q, there is an a in the convex hull of u, v, w for which $x_1 \leq a_1$, $x_2 \geq a_2$ and therefore the set of lines with positive slope which separate x from u, v, w is empty.

(b) Case: $\bar{a} \notin \text{convhull}(u, v, w)$: For any x in the upper left quadrant Q which is not in the yellow region, there is an a in the convex hull of u, v, w for which $x_1 \leq a_1$, $x_2 \geq a_2$ and therefore the set of lines with positive slope which separate x from u, v, w is empty. If x is in the yellow region, the line through z_r and z_l has positive slope and separates x from u, v, w.

Figure 7 Examples of configurations of points for Proposition 8, case 1. In both figures, we may use vertical and horizontal lines through $\bar{a}$ to partition the plane into four quadrants. For points x to the right of the vertical line through $\bar{a}$, we may rotate that line slightly clockwise around $\bar{a}$ to obtain a line with positive slope separating x from u, v, w. For points x below the horizontal line through $\bar{a}$, we may rotate that line slightly counterclockwise around $\bar{a}$ to obtain a line with positive slope separating x from u, v, w. For points x such that $x_1 \leq \bar{a}_1$ and $x_2 \geq \bar{a}_1$, the cases $\bar{a} \in \text{convhull}(u, v, w)$ and $\bar{a} \notin \text{convhull}(u, v, w)$ must be studied separately.

Let us assume there is a point $a = (a_1, a_2)$ in the convex hull of u, v, w such that $x_1 \leq a_1$ and $x_2 \geq a_2$. As a is in the convex hull of u, v, w, if there were a line $L : y_2 = my_1 + q$ with slope $m > 0$ strictly separating x from u, v, w, then L would also separate x from a. So, $a_2 \geq ma_1 + q$ while $x_2 < mx_1 + q$. Now, the assumptions $x_1 \leq a_1$ and $x_2 \geq a_2$ imply $x_2 \geq mx_1 + q$, i.e. that x lies above L, yielding a contradiction. Therefore, the existence of

such point a implies that the subset of the lines of positive slope that do not contain x in $E(A(u,v,w,x))$ is empty.

Vice versa, let us assume that there is no point $a=(a_1,a_2)$ in the convex hull of u, v, w such that $x_1 \leq a_1$ and $x_2 \geq a_2$. Let $\bar{a}_1 \in \mathbb{R}$ be the maximum abscissa of points a in the convex hull of u,v,w, and $\bar{a}_2 \in \mathbb{R}$ their minimum ordinate. Set $\bar{a}=(\bar{a}_1,\bar{a}_2) \in \mathbb{R}^2$.

In the case when $\bar{a}$ belongs to the convex hull of u,v,w, by assumption we have $x_1 > \bar{a}_1$ or $x_2 < \bar{a}_2$. If $x_1 > \bar{a}_1$, let us consider the vertical line L' through $\bar{a}$ so that x is strictly to the right of L'. No point of L' below $\bar{a}$ and no point to the right of L' belong to the convex hull of u,v,w. Thus, slightly rotating clockwise the line L' about $\bar{a}$, we obtain a valid line $L \in E(A(u,v,w,x))$. Similarly, if $x_2 < \bar{a}_2$, let us consider the horizontal line L'' through $\bar{a}$ so that x is strictly below L''. No point of L'' to the right of $\bar{a}$ and no point below L'' belongs to the convex hull of u,v,w. Thus, slightly rotating counter-clockwise the line L'' about $\bar{a}$, we obtain a valid line $L \in E(A(u,v,w,x))$.

In the case when $\bar{a}$ does not belong to the convex hull of u,v,w, there are among u,v,w a rightmost point z_r with abscissa $\bar{a}_1$ and a different lowest point z_ℓ with ordinate $\bar{a}_2$. The line through z_r and z_ℓ has positive slope and all the points $a=(a_1,a_2)$ in the segment $z_r z_\ell$ belong to the convex hull of u,v,w, so $x_1 > a_1$ or $x_2 < a_2$. In particular, if $x_1 > \bar{a}_1$, then $x_1 > a_1$ for every point in the segment $z_r z_\ell$. So, we can take as L' the vertical line through z_r. As before, slightly rotating L' about z^r clockwise we get a line L as desired. Otherwise, if $x_2 < \bar{a}_2$, then $x_2 < a_2$ for every point in the segment $z_r z_\ell$. So, we can take as L'' the horizontal line through z_ℓ. As before slightly rotating L'' about z^ℓ counterclockwise we get a line L as desired. Finally, if $x_1 \leq \bar{a}_1$ and $x_2 \geq \bar{a}_2$, let us consider the line L through z^r and z^ℓ and show that it belongs to $E(A(u,v,w,x))$. Indeed, by construction, L has positive slope, the points u,v,w are to the left of or on L so they push right onto L and x is strictly on the right of L because $x_1 \leq \bar{a}_1$, $x_2 \geq \bar{a}_2$, and, for all the points $a=(a_1,a_2)$ of the segment $z_r z_\ell$, $x_1 > a_1$ or $x_2 < a_2$. Hence, the line L belongs to $E(A(u,v,w,x))$.

It remains to prove the third statement (the fourth being similar). The proof is similar to that of the first statement.

Let us assume there is a point $a=(a_1,a_2)$ in the convex hull of u,v,w such that $x_1 \leq a_1$, $x_2 \geq a_2$, and $x \neq a$, i.e. $x_1 < a_1$ and $x_2 > a_2$. As a is in the convex hull of u,v,w, if there were a line $L: y_2 = my_1 + q$ with slope $m > 0$ separating x from u,v,w, then L would also separate x from a. So, $a_2 \geq ma_1 + q$ while $x_2 \leq mx_1 + q$. Now, the assumptions $x_1 < a_1$ and $x_2 > a_2$ imply $x_2 > mx_1 + q$, a contradiction.

To prove the opposite implication for statement 3, we notice that we may use an identical argument as in statement 1, except that strict inequalities become inclusive inequalities. However, all of the results still hold, so that the opposite implication in statement 3 holds.

◀

Proof of Proposition 9. We will only prove case *1*, as case *2* is completely similar.

Let us first assume that $E(A(u,v,w,x))$ contains a line of positive slope L with $\Delta_L(u,v,w,x;\delta,\eta) = 0$. By [2, Lemma 4.1], such line L must pass through a point $\omega=(\omega_1,\omega_2)$ determined by either Equation (1) or Equation (6). So, also $E(A(u,v,w,\omega))$ with $A_\omega = \{1,2\}$ is non-empty. In either Equations (1) and (6), $\omega_1 = x_1$, so that the assumption $A_x = \{1\}$, equivalent to $x \notin L$ and x pushing upwards, implies that $\omega_2 > x_2$. Moreover, the third statement of Proposition 8 applied to the points u,v,w,ω, implies that there is no point $a=(a_1,a_2)$ in the convex hull of u,v,w for which $\omega_1 \leq a_1$, $\omega_2 \geq a_2$, and $\omega \neq a$.

Vice versa, we must show that among the lines L of $E(A(u,v,w,x))$, which by assumption is non-empty, there is at least one line for which $\Delta_L(u,v,w,x;\delta,\eta) = 0$. As ω is obtained by Equation (1) or Equation (6), we have $\omega_1 = x_1$. Because we are assuming that there

is no point $a = (a_1, a_2)$ in the convex hull of u, v, w with $\omega_1 \leq a_1$, $\omega_2 \geq a_2$, and $\omega \neq a$, the third statement of Proposition 8 implies that $E(A(u, v, w, \omega))$ with $2 \in A_u \cap A_v \cap A_w$ and $A_\omega = \{1, 2\}$, is non-empty. Let L' be a line in $E(A(u, v, w, \omega))$. From the fact that L' passes through ω, $\omega_1 = x_1$, and we are assuming that $\omega_2 > x_2$, it follows that $A_x^{L'} = \{1\}$. So, $L' \in E(A(u, v, w, x))$.

Recall that Equation (1) (resp. Equation (6)) for ω are obtained by imposing $\Delta_L(u, v, w, x; \delta, \eta) = 0$ for $L \in E(A(u, v, w, x))$, implying that $\Delta_{L'}((u, v, w, x; \delta, \eta) = 0$. ◀

Proof of Proposition 10. We only prove the first equivalence, the second one being analogous. Let us consider the sets

$$A := \{a \in \mathbb{R}^2 \mid a \in \text{convhull}(u, v, w), \omega_1 \leq a_1, \omega_2 \geq a_2, \text{ and } a \neq \omega\}$$

and

$$B := \{a \in \mathbb{R}^2 \mid a \in \text{convhull}(u, v, w), \omega_1 \leq a_1, \omega_2 \geq a_2\}.$$

Clearly, B contains A and either $B = A \cup \{\omega\}$ or $B = A$. Thus, $A = \emptyset$ if and only if $B = \emptyset$ or $B = \{\omega\}$. In the first case, Algorithm 1 can be used on the quadruple u, v, w, ω to determine if B is empty: $B = \emptyset$ if and only if CheckPts$(u, v, w, \omega, \text{Below}) =$ True. In the second case, $B = \{\omega\}$ if and only if ω is a point on the boundary of the convex hull of u, v, w. In this case, Algorithm 1 will return False for u, v, w, ω. ◀

Proof of Proposition 11. We know that l.u.b.$(w, x) = x$ if and only if $x_1 \geq w_1$ and $x_2 \geq w_2$. In this case, it must also be true that $\text{push}_L(x)_1 \geq \text{push}_L(w)_1$ and $\text{push}_L(x)_2 \geq \text{push}_L(w)_2$. By [2, Equation (5)], this ensures that $p_L(x) \geq p_L(w)$. If l.u.b.$(w, x) = w$, the above inequalities are swapped, and we obtain $p_L(x) \leq p_L(w)$.

If l.u.b.(w, x) is neither x nor w, then the sign of $p_L(x) - p_L(w)$ depends on how L separates w, x, and l.u.b.(w, x). If L separates x and l.u.b.(w, x) from w, then $\text{push}_L(x) = \text{push}_L(\text{l.u.b.}(w, x))$. It is again the case that l.u.b.$(w, x)_1 \geq w_1$ and l.u.b.$(w, x)_2 \geq w_2$, so that $\text{push}_L(\text{l.u.b.}(w, x))_1 \geq \text{push}_L(w)_1$ and $\text{push}_L(\text{l.u.b.}(w, x))_2 \geq \text{push}_L(w)_2$. Since we have $\text{push}_L(x) = \text{push}_L(\text{l.u.b.}(w, x))$, we may again use [2, Equation (5)] to see that $p_L(x) \geq p_L(w)$. If L separates w and l.u.b.(w, x) from x, then the above inequalities are swapped, and we obtain $p_L(x) \leq p_L(w)$. ◀

Proof of Proposition 12. *(iii)* implies *(ii)*:

Assume that there are a point $a = (a_1, a_2)$ in the convex hull of u, v and a point $b = (b_1, b_2)$ in the convex hull of x, w such that $(b_1 \geq a_1) \wedge (b_2 \leq a_2)$. If there is a line $L : y_2 = my_1 + q$ which separates u, v from w, x with u, v pushing right and x, w pushing up, then this line also strictly separates a from b, with a pushing right, and b pushing up to L. So, $a_2 > ma_1 + q$ and $b_2 < mb_1 + q$. However, the assumption that $(b_1 \leq a_1) \wedge (b_2 \geq a_2)$ implies that

$$b_2 < mb_1 + q \leq ma_1 + q < a_2 \leq b_2,$$

which is a contradiction. Hence, the set of lines $E(A(u, v, w, x))$ is empty.

(ii) implies *(iii)*:

We show the contrapositive statement: If, for every point $a = (a_1, a_2)$ on $\overline{uv}$ and every point $b = (b_1, b_2)$ on $\overline{xw}$, we have that

$$(b_1 > a_1) \vee (b_2 < a_2)\,, \tag{10}$$

then there exists a line L in the set $E(A(u, v, w, x))$, i.e. a line with positive slope that separates u and v from x and w with $A_u^L = A_v^L = \{2\}$ and $A_x^L = A_w^L = \{1\}$.

We start noting that a line $L : y_2 = my_1 + q$ separates u and v from x and w with $A_u^L = A_v^L = \{2\}$ and $A_x^L = A_w^L = \{1\}$ exactly when, for all $a = (a_1, a_2) \in \overline{uv}$ and all $b = (b_1, b_2) \in \overline{xw}$, the parameters m and q satisfy the following system of inequalities:

$$\begin{cases} a_2 > ma_1 + q \\ b_2 < mb_1 + q\,, \end{cases} \tag{11}$$

or equivalently $a_2 - ma_1 > q > b_2 - mb_1$.

For a given pair of points a, b, this system can be solved for m and q if and only if there exists an m such that $a_2 - ma_1 > b_2 - mb_1$, or equivalently $a_2 - b_2 > m(a_1 - b_1)$. So there exists a line with positive slope that separates a from b with $A_a^L = \{2\}$ and $A_b^L = \{1\}$ if and only if there is a positive m such that $a_2 - b_2 > m(a_1 - b_1)$. But this is the case exactly when we have $(b_1 > a_1) \vee (b_2 < a_2)$, the assumption we are making for any $a = (a_1, a_2) \in \overline{uv}$ and $b = (b_1, b_2) \in \overline{xw}$. Therefore, we now know that, for a given pair of points $a = (a_1, a_2) \in \overline{uv}$ and $b = (b_1, b_2) \in \overline{xw}$, we can always find a line $L : y_2 = my_1 + q$ with positive slope that separates a from b with $A_a^L = \{2\}$ and $A_b^L = \{1\}$.

Next, note that for a given pair of points $a = (a_1, a_2)$ and $b = (b_1, b_2)$ that satisfies the condition that $(b_1 > a_1) \vee (b_2 < a_2)$, we have exactly one of the following three scenarios:

- Scenario I: $b_1 > a_1$ and $b_2 > a_2$
- Scenario II: $b_1 > a_1$ and $b_2 < a_2$
- Scenario III: $b_1 < a_1$ and $b_2 < a_2$

The scenario when $b_1 < a_1$ and $b_2 > a_2$ is excluded by the assumption. To find line parameters m and q that solve the system of inequalities (11) for a given pair of points $a = (a_1, a_2)$ and $b = (b_1, b_2)$, it may be helpful to visualise the lines $q = -b_1m + b_2$ and $q = -a_1m + a_2$ in the (m, q)-parameter space (see Figure 8). Note that in scenarios I and II, one can choose m arbitrarily large, while in scenarios II and III, one can choose m to be a positive number arbitrarily close to 0.

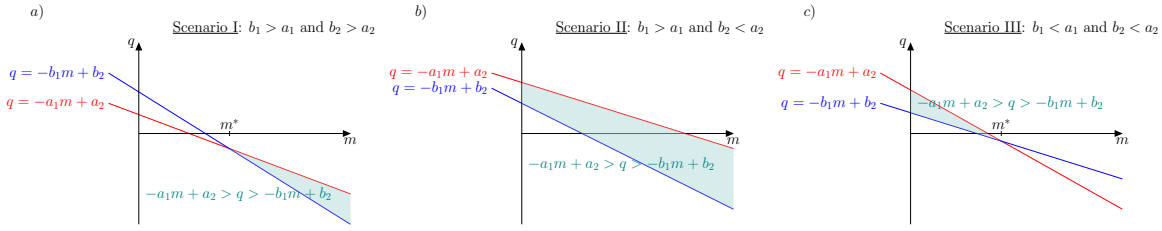


Figure 8 The lines $q = -a_1m + a_2$ (red) and $q = -b_1m + b_2$ (blue) and the regions in the (m, q)-parameter space corresponding to lines of positive slope that separate $a = (a_1, a_2)$ from $b = (b_1, b_2)$ with $A_a^L = \{2\}$ and $A_b^L = \{1\}$ (shaded in teal).

To find a line of positive slope that separates u and v from x and w with $A_u^L = A_v^L = \{2\}$ and $A_x^L = A_w^L = \{1\}$, we need to find a positive m and a q that solve 11 for all $a = (a_1, a_2) \in \overline{uv}$ and $b = (b_1, b_2) \in \overline{xw}$. We will show below that there is always one endpoint of the segment $\overline{uv}$, one endpoint of the segment $\overline{xw}$, and a line with positive slope that separates these endpoints from each other in the desired way and whose slope guarantees that it in fact separates all of $\overline{uv}$ from all of $\overline{xw}$ in the desired way. The existence of such a line proves that $E(A(u, v, w, x))$ is non-empty.

First, we define

$$m_a := \begin{cases} \text{the slope of the line segment } \overline{uv} & \text{if } u \neq v \\ 0 & \text{if } u = v \ , \end{cases} \tag{12}$$

and

$$m_b := \begin{cases} \text{the slope of the line segment } \overline{xw} & \text{if } x \neq w \\ 0 & \text{if } x = w \ . \end{cases} \tag{13}$$

We also define the following notation for the endpoints of the line segments $\overline{uv}$ and $\overline{uv}$:

- Let $\{a', a''\} = \{u, v\}$ such that either $a_1'' < a_1'$ or $(a_1' = a_1'') \wedge (a_2'' \leq a_2')$ or $(a_1' = a_1'') \wedge (a_2'' = a_2')$, and
- let $\{b', b''\} = \{x, w\}$ such that either $b_1' < b_1''$ or $(b_1' = b_1'') \wedge (b_2' \leq b_2'')$ or $(b_1' = b_1'') \wedge (b_2'' = b_2')$.

With this notation, we can observe that the following four statements hold.

(S_1) Any line L with positive slope $m \geq m_a$ and with $A^L_{a'} = \{2\}$ is such that $A^L_a = \{2\}$ for any $a \in \overline{uv}$.
(S_2) Any line L with positive slope $m \leq m_a$ and with $A^L_{a''} = \{2\}$ is such that $A^L_a = \{2\}$ for any $a \in \overline{uv}$.
(S_3) Any line L with positive slope $m \geq m_b$ and with $A^L_{b'} = \{1\}$ is such that $A^L_b = \{1\}$ for any $b \in \overline{xw}$.
(S_4) Any line L with positive slope $m \leq m_b$ and with $A^L_{b''} = \{1\}$ is such that $A^L_b = \{1\}$ for any $b \in \overline{xw}$.

Given u, v, x and w, we can distinguish four cases based on the sign of the slopes of the line segments $\overline{uv}$ and $\overline{xw}$ (see Figure 9).

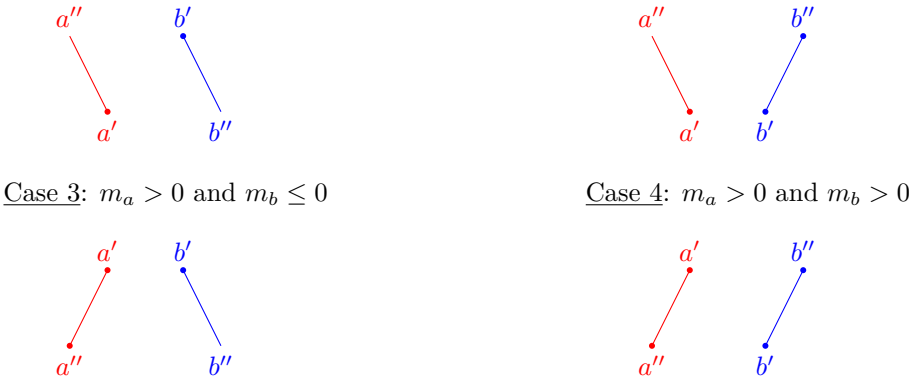


Figure 9 Illustration of slopes of the line segments $\overline{uv}$ and $\overline{xw}$ in the four different cases we consider. Note that these illustrations do not reflect all possible constellations. In particular, they do not reflect the cases where either $u = v$ or $x = w$.

In the case of both m_a and m_b being positive, we distinguish further two subcases:

- Case 4a): $m_a \geq m_b$

- Case 4b): $m_a < m_b$

In each of these cases, we next show that there is an endpoint of the segment $\overline{uv}$ and an endpoint of the segment $\overline{xw}$ and a line L with positive slope separating these two endpoints in the desired way which in fact separates all of $\overline{uv}$ from all of $\overline{xw}$ in the desired way, that is with $A_u^L = A_v^L = \{2\}$ and $A_x^L = A_w^L = \{1\}$.

Case 1: $m_a \leq 0$ and $m_b \leq 0$

As observed above, by applying the assumption of Equation (10) to a' to b' (i.e. $(b'_1 > a'_1) \vee (b'_2 < a'_2)$), we can find a line L of positive slope m with $A_{a'}^L = \{2\}$ and $A_{b'}^L = \{1\}$. As any positive slope m is greater than m_a and m_b, in this case, the line L is such that $A_a^L = \{2\}$ for all $a \in \overline{uv}$ by statement (S_1) above, and such that $A_b^L = \{1\}$ for all $b \in \overline{xw}$ by statement (S_3) above. In other words, $L \in E(A(u, v, w, x))$.

Case 2: $m_a \leq 0$ and $m_b > 0$.

If a' and b' give scenario I or II, then we can find a line L with positive slope $m \geq m_b > 0 \geq m_a$ and with $A_{a'}^L = \{2\}$ and $A_{b'}^L = \{1\}$, as the range of possible slopes is not bounded above in these scenarios (cf. Figure 8(a-b). By statements (S_1) and (S_3) above, L is such that $A_a^L = \{2\}$ for all $a \in \overline{uv}$ and such that $A_b^L = \{1\}$ for all $b \in \overline{xw}$.

Similarly, if a' and b'' give scenario II or III, we can find a line L with positive slope m such that $m_a < m \leq m_b$ and with $A_{a'}^L = \{2\}$ and $A_{b''}^L = \{1\}$, and this line L is such that $A_a^L = \{2\}$ for all $a \in \overline{uv}$ and such that $A_b^L = \{1\}$ for all $b \in \overline{xw}$ by statements (S_1) and (S_4) above.

Finally, if a' and b' give scenario III while a' and b'' give scenario I, we first note that the following inequalities hold:

$$\begin{aligned} &b'_1 < a'_1 < b''_1, \\ &b'_2 < a'_2 < b''_2 \,. \end{aligned} \tag{14}$$

These inequalities together with the condition on the relative position of a' to any point $b \in \overline{xw}$, i.e. $(b_1 > a'_1) \vee (b_2 < a'_2)$, imply that the point a' lies in the interior of the triangle with vertices $(b'_1, b'_2), (b''_1, b''_2)$ and (b'_1, b''_2) and that the slope of the line segment $\overline{a'b'}$ is larger than m_b (cf. Figure 10a)). Now note that the m-coordinate of the intersection between the lines $q = -b'_1 m + b'_2$ and $q = -a'_1 m + a'_2$ in the (m, q)-parameter space is $m^* = \frac{a'_2 - b'_2}{a'_1 - b'_1}$, i.e. the slope of the line segment $\overline{a'b'}$. So by our observation above, we know that $m^* > m_b$, and, as a' and b' give scenario III, we can see from Figure 8c) that there exists a line L of positive slope $m \geq m_b$ that separates a' from b' with $A_{a'}^L = \{2\}$, $A_{b'}^L = \{1\}$. Thus, by statements (S_1) and (S_3) we have that $L \in E(A(u, v, w, x))$. Note that one can argue similarly to show that there exists a line L' of positive slope $m' \leq m_b$ that separates a' from b'' with $A_{a'}^{L'} = \{2\}$, $A_{b''}^{L'} = \{1\}$, and that we have that $L' \in E(A(u, v, w, x))$ by statements (S_1) and (S_4).

Case 3: $m_a > 0$ and $m_b \leq 0$.

This case is analogous to Case 2. There is a line L with positive $m \leq m_a$ such that $A_{a''}^L = \{2\}$, $A_{b'}^L = \{1\}$, and by statements (S_2) and (S_3) we have that $L \in E(A(u, v, w, x))$, and there is a line L' with slope $m' \geq m_a$ such that $A_{a'}^{L'} = \{2\}$, $A_{b'}^{L'} = \{1\}$, and by statements (S_1) and (S_3) we have that $L' \in E(A(u, v, w, x))$.

Case 4: $m_a > 0$ and $m_b > 0$.

If a'' and b'' give scenario II or III, then we can find a line L with positive slope $m \leq m_a, m_b$ such that $A_{a''}^L = \{2\}$, $A_{b''}^L = \{1\}$, and by statements (S_2) and (S_4) we have that $L \in E(A(u, v, w, x))$ (as m can be chosen arbitrarily close to 0 in these scenarios).

If a' and b' give scenario I or II, then we can find a line L with slope $m \geq m_a, m_b$ such that $A_{a'}^L = \{2\}$, $A_{b'}^L = \{1\}$, and by statements (S_1) and (S_3) we have that $L \in E(A(u, v, w, x))$ (as m can be chosen arbitrarily large in these scenarios).

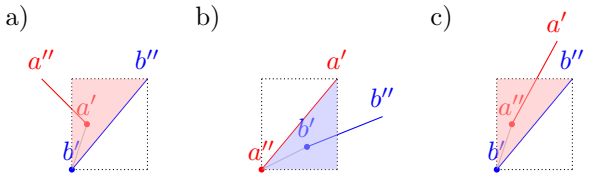


Figure 10 Illustration of constellations of the points a', a'', b' and b'' in subcases of case 2) and 4a). Panel a) shows the relative position of a' with respect to $\overline{b'b''}$ when a' and b' give scenario III while a' and b'' give scenario I in case 2. Panel b) shows the relative position of b' with respect to $\overline{a''a'}$ when a'' and b' give scenario I, and panel c) shows the relative position of a'' with respect to $\overline{b'b''}$ when a'' and b' give scenario III.

If a'' and b'' give scenario I and a' and b' give scenario III, we first note that the following inequalities hold:

$$a''_1 < b''_1, \quad a''_2 < b''_2, \quad a'_1 > b'_1, \quad a'_2 > b'_2 \ . \tag{15}$$

We now consider the subcases 4a) and 4b) separately:

Case 4a): $m_a \geq m_b > 0$. We distinguish three further subcases based on the relative position of a'' to b':

- a'' and b' give scenario I,
- a'' and b' give scenario II, and
- a'' and b' give scenario III.

In the case where a'' and b' give scenario I, we have that

$$a''_1 < b'_1, \text{ and } a''_2 < b'_2 \ . \tag{16}$$

Combining Equation (16) and Equation (15) gives

$$a''_1 < b'_1 < a'_1 \text{ and } a''_2 < b'_2 < a'_2 \ , \tag{17}$$

and together with our assumption on the relative position of b' to any point $a \in \overline{uv}$, namely that $(b'_1 > a_1) \vee (b'_2 < a_2)$, this implies that b' lies in the interior of the triangle with vertices $(a''_1, a''_2), (a'_1, a''_2)$ and (a'_1, a'_2), which in turn implies that the slope of the line segment $\overline{a''b'}$ is smaller than m_a (cf. Figure 10b)). But this guarantees the existence of a line L separating a'' from b' with $A^L_{a''} = \{2\}$ and $A^L_{b'} = \{1\}$ whose slope m is such that $m_b \leq m \leq m_a$: The m-coordinate of the intersection between the lines $q = -b'_1 m + b'_2$ and $q = -a''_1 m + a''_2$ in the (m, q)-parameter space is $m^* = \frac{a''_2 - b'_2}{a''_1 - b'_1}$, i.e. the slope of the line segment $\overline{a''b'}$. So by our observation above, we know that $m^* < m_a$, and, as a'' and b' are in scenario I, we can see from Figure 8a) that there exists a line L separating a'' from b' with $A^L_{a''} = \{2\}$ and $A^L_{b'} = \{1\}$ whose slope m is such that $m_b \leq m \leq m_a$.

If, on the other hand, a'' and b' give scenario II, we can see in Figure 8b) that, for any positive m, there exists a line L with slope m that separates a'' from b' with $A^L_{a''} = \{2\}$ and $A^L_{b'} = \{1\}$. In particular, we can find a line L separating a'' from b' with $A^L_{a''} = \{2\}$ and $A^L_{b'} = \{1\}$ whose slope m is such that $m_b \leq m \leq m_a$.

Finally, if a'' and b' give scenario III, we have that

$$a''_1 > b'_1 \text{ and } a''_2 > b'_2 \ . \tag{18}$$

Combining Equation (18) and Equation (15) gives

$$b'_1 < a''_1 < b''_1 \text{ and } b'_2 < a''_2 < b''_2 \ , \tag{19}$$

and together with our assumption on the relative position of a'' to any point $b \in \overline{xw}$, namely that $(b_1 > a''_1) \vee (b_2 < a''_2)$, this implies that a'' lies in the interior of the triangle with vertices $(b'_1, b'_2), (b''_1, b''_2)$, and (b'_1, b''_2), which in turn implies that the slope of the line segment $\overline{a''b'}$ is greater than m_b (cf. Figure 10c)). But this guarantees the existence of a line L separating a'' from b' with $A^L_{a''} = \{2\}$ and $A^L_{b'} = \{1\}$ whose slope m is such that $m_b \leq m \leq m_a$: Just as in scenario I, the m-coordinate of the intersection between the lines $q = -b'_1 m + b'_2$ and $q = -a''_1 m + a''_2$ in the (m, q)-parameter space is $m^* = \frac{a''_2 - b'_2}{a''_1 - b'_1}$, i.e. the slope of the line segment $\overline{a''b'}$. So by our observation above, we know that $m^* > m_b$, and, as a'' and b' are in scenario III, we can see from Figure 8c) that there exists a line L separating a'' from b' with $A^L_{a''} = \{2\}$ and $A^L_{b'} = \{1\}$ whose slope m is such that $m_b \leq m \leq m_a$.

So in any subcase, whether a'' and b' give scenario I, II, or III, we know that there is a line L separating a'' from b' with $A^L_{a''} = \{2\}$ and $= A^L_{b'} = \{1\}$ whose slope m is such that $m_b \leq m \leq m_a$. By statements (S_2) and (S_3) above, L is such that $A^L_a = \{2\}$ and $= A^L_b = \{1\}$ for any $a \in \overline{uv}$ and $b \in \overline{xw}$. In other words, L separates u and v from x and w in the desired way, and so $E(A(u, v, w, x))$ is non-empty.

Case 4b): $m_b > m_a > 0$. This case is analogous to Case 4a). We can find a line L with slope m such that $m_a \leq m \leq m_b$ and with $A^L_{a'} = \{2\}$ and $A^L_{b''} = \{1\}$, and by statements (S_1) and (S_4) we have that L is such that $A^L_a = \{2\}$ and $= A^L_b = \{1\}$ for any $a \in \overline{uv}$ and $b \in \overline{xw}$. In other words, $L \in E(A(u, v, w, x))$ and hence $E(A(u, v, w, x))$ is non-empty.

In summary, we have found that in any case (Case 1, 2, 3, 4a), and 4b)) it is possible to find a line in the set $E(A(u, v, w, x))$, so this set is non-empty, and we have proven that not-(iii) implies not-(ii), and, equivalently, that (ii) implies (iii).

(iv) implies (iii):
If it is the case that x is in Q, then by definition of Q, $x_1 \leq a_1$ and $x_2 \geq a_2$ for some $a \in \overline{uv}$. Thus, x and a satisfy the conditions given in (2). If it is otherwise the case $w \in Q$ or u or $v \in R$, similar pairs which satisfy the conditions of (2) may be found with w, u or v, respectively.

(iii) implies (i):
Note that $\overline{uv} \in Q$ and $\overline{xw} \in R$. If additionally we assume that there is some point $a = (a_1, a_2)$ on $\overline{uv}$ and a point $b = (b_1, b_2)$ on $\overline{xw}$ such that $(b_1 \leq a_1) \wedge (b_2 \geq a_2)$, then it is the case that $a \in R$ and $b \in Q$. Thus, $a, b \in R \cap Q$, so clearly $R \cap Q \neq \emptyset$.

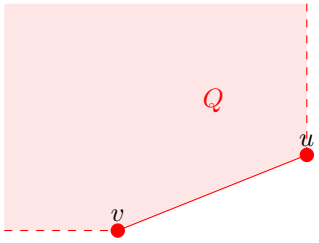
(i) implies (iv):
To fix the ideas, we assume wlog $(u_2 > v_2) \vee ((u_2 = v_2) \wedge (u_1 < v_1))$, and $(w_1 > x_1) \vee ((w_1 = x_1) \wedge (w_2 < x_2))$.

By definition of Q, if $m_a \leq 0$ or $m_a = \infty$ (cf. Equation (12)), then $Q = \{(p_1, p_2) \in \mathbb{R}^2 | p_1 \leq v_1 \text{ and } p_2 \geq v_2\}$. If $\infty \neq m_a > 0$, then Q is the upper-region in $\mathbb{R}^2$ bounded by

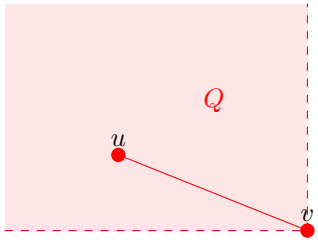
$$\{(p_1, v_2) \in \mathbb{R}^2 | p_1 \leq v_1\} \cup \overline{uv} \cup \{(u_1, p_2) \in \mathbb{R}^2 | p_2 \geq u_2\}.$$

Similarly, if $m_b \leq 0$ or $m_b = \infty$ (cf. Equation (13)) , then $R = \{(p_1, p_2) \in \mathbb{R}^2 | p_1 \geq w_1 \text{ and } p_2 \leq w_2\}$. If $\infty \neq m_a > 0$, then R is the lower-region in $\mathbb{R}^2$ bounded by

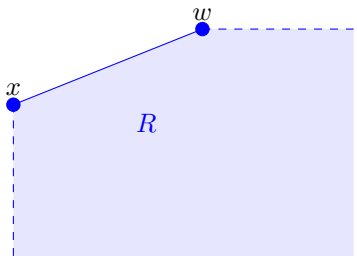
$$\{(x_1, p_2) \in \mathbb{R}^2 | p_2 \leq x_2\} \cup \overline{xw} \cup \{(p_1, w_2) \in \mathbb{R}^2 | p_1 \geq w_1\}.$$



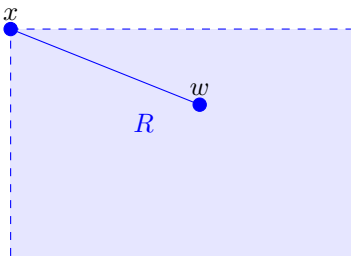
(a) The region Q, when $\overline{uv}$ has positive slope, is shaded in red.



(b) The region Q, when $\overline{uv}$ has negative slope, is shaded in red.



(c) The region R, when $\overline{xw}$ has positive slope, is shaded in blue.



(d) The region R, when $\overline{xw}$ has negative slope, is shaded in blue.

Figure 11 The regions Q and R defined in Proposition 12

With this in mind, assume that $R \cap Q \neq \emptyset$. If it is the case that $m_b \leq 0$ or $m_b = \infty$, then from the assumption $x_1 \leq w_1$ it follows that $x_2 \geq w_2$, and thus we have $x \in Q$. The case when $m_a \leq 0$ or $m_a = \infty$ is analogous.

If both $\infty \neq m_a > 0$ and $\infty \neq m_b > 0$, it must be the case that at least one of the boundary components of R intersects Q and vice versa.

- If $\{(x_1, p_2) \in \mathbb{R}^2 | p_2 \leq x_2\}$ intersects Q, then $x \in Q$.
- If $\{(p_1, w_2) \in \mathbb{R}^2 | p_1 \geq w_2\}$ intersects Q, then $w \in Q$.
- If neither of the unbounded components of the boundary of R intersects Q, it must be the case that at least one of the unbounded boundary components of Q intersects R. Using the same justification as in the previous two points, this means that either u or v is in R.

◀

Proof of Proposition 13. If the equivalence class $E(A(u, v, w, x))$ contains some lines L for which $\Delta_L(u, v, w, x; \delta, \eta) = 0$, then by [2, Lemma 4.1] it passes through ω as given in Equation (3). This implies that L has slope $m = \frac{\delta(w_1 - x_1)}{\eta(u_2 - v_2)}$, which is positive by the assumption $p_L(u) > p_L(v)$ and $p_L(w) > p_L(x)$. As for the y-intercept q of L, because L keeps u, v on the left and x, w on the right, there must exist $q \in \mathbb{R}$ such that $\max\{x_2 - mx_1, w_2 - mw_1\} < q < \min\{u_2 - mu_1, v_2 - mv_1\}$, proving Equation (9). Vice versa, we must show that among the lines L of $E(A(u, v, w, x))$, which by assumption is non-empty, there is at least one line for which $\Delta_L(u, v, w, x; \delta, \eta) = 0$. If $\max\{x_2 - mx_1, w_2 - mw_1\} < \min\{u_2 - mu_1, v_2 - mv_1\}$ then taking q between these two values, the line $L : y = mx + q$ is such that it passes through ω and correctly separates u, v from x, w. As $\omega = [0 : \delta(w_1 - x_1) : \eta(u_2 - v_2)]$ has been obtained from Equation (6) imposing $\Delta_L(u, v, w, x; \delta, \eta) = 0$, we can conclude. ◀

Proof of Proposition 14. The first two cases are self-evident. Suppose then that $x, v \notin Q_2$, $u, w \notin Q_4$, and at least one of x, v, u, w is in Q_1 or Q_3. Notice that if $x \in Q_4$, then x will push up to any line through ω with positive slope - the same is true for $v \in Q_4$. Similarly, if $u \in Q_2$, then u will push right to any line through ω with positive slope, and the same is true for $w \in Q_2$. So, we only need to consider the points c in $\{x, v, u, w\} \cap (Q_1 \cup Q_3)$, which by assumption is non-empty. If $c \in Q_1 \cap P$, then a line L through ω such that $A_c = \{1\}$ exists if and only if its slope m is such that $m > m_c$, where m_c is the slope through ω and c. If $c \in Q_3 \cap P$, then the condition is $m < m_c$. Analogous conditions apply for $A_c = \{2\}$, $c \in Q_1 \cap S$ and $c \in Q_3 \cap S$.

◀

B Pseudo-code

Algorithm 1 Function to assess if, for four points x, u, v, w, there is a line L with positive slope separating x from u, v, w.

Input: u, v, w, x, Position
Output: True if the configuration is good and False otherwise.

function CheckPts(u, v, w, x, Position)

  if Position == Below **then** ▷ case x strictly below the line and u, v, w above the line
    $z_1 \leftarrow \max(u_1, v_1, w_1)$
    $z_2 \leftarrow \min(u_2, v_2, w_2)$
    $z \leftarrow (z_1, z_2)$
    $Q \leftarrow$ the closed upper-left quadrant with origin at z
  end if

  if Position == Above **then** ▷ case x strictly above the line and u, v, w below the line
    $z_1 \leftarrow \min(u_1, v_1, w_1)$
    $z_2 \leftarrow \max(u_2, v_2, w_2)$
    $z \leftarrow (z_1, z_2)$
    $Q \leftarrow$ the closed lower-right quadrant with origin at z
  end if

  if $x \in Q$ **then**
    if $x \in$ convhull(u, v, w) **then**
      return False
    else if $x \in$ convhull(u, v, w, z) **then**
      return True
    else
      return False
    end if
  else
    return True
  end if
end function

Algorithm 2 Functions to assess if ω satisfies the conditions of Proposition 9 and is not superfluous.

Input: ω, u, v, w, x, Position. **Output:** True, if ω is correct and not superfluous, otherwise False.

```
1: function CheckOmega(ω, u, v, w, x, Position)
2:
3:     if Position == Below then                ▷ case x strictly below the line and u, v, w above the line
4:         if ω_2 ≤ x_2 then
5:             return False
6:         end if
7:
8:     else if Position == Above then           ▷ case x strictly above the line and u, v, w below the line
9:         if ω_1 ≤ x_1 then
10:            return False
11:        end if
12:    end if
13:
14:    if CheckPts(u, v, w, ω, Position) == False then   ▷ Both cases x strictly above or below the line
15:        return False
16:    else
17:        return True
18:    end if
19: end function
```

Algorithm 3 An algorithm to compute switch points ω from a set of four points c_1, c_2, c_3, c_4 so that three points are one side of a line through ω and the remaining point is strictly on the other side.

Input: $c_1, c_2, c_3, c_4 \in C_M \sqcup C_N$. **Output:** The partial list of switch points ω generated by Case 2 in [2, Lemma 4.1]
Ensure: there are 3 distinct points among $\{c_1, c_2, c_3, c_4\}$

function OmegaPoints3v1(c_1, c_2, c_3, c_4)
  $\Omega \leftarrow \emptyset$
  CritPts $\leftarrow \{c_1, c_2, c_3, c_4\}$
  for c in CritPts **do**
    if check_c_distinct(c, CritPts\{c}) **then** ▷ x must be different from u, v, w
      $x \leftarrow c$
    else
      continue
    end if
    for d in CritPts.remove(x) **do** ▷ Note: w therefore is distinct from x
      $w \leftarrow d$
      if both elements of CritPts\{x,w} are the same **then continue** ▷ The crit vals assigned to u and v cannot be the same.
      end if
      $u \leftarrow$ an element of CritPts\{x,w}
      $v \leftarrow$ CritPts\{x,w,u}
      if x.parent() == w.parent() **then** ▷ determining η
        $\eta \leftarrow 2$
      else
        $\eta \leftarrow 1$
      end if
      if u.parent() == v.parent() **then** ▷ determining δ
        $\delta \leftarrow 2$
      else
        $\delta \leftarrow 1$
      end if
      ▷ Case when x pushes up, u, v, w push right.
      if CheckPts(CritPts, Below) **then** ▷ Checking if the configuration is possible with x below the line and u, v, w above the line
        if $u_2 == v_2$ **then continue** ▷ This matching has null cost.
        end if
        if $u_2 < v_2$ **then** ▷ ensuring $p_L(u) > p_L(v)$
          Exchange u, v.
        end if
        if l.u.b.$(x, w) = w$ **then** ▷ Case where $p_L(w) \geq p_L(x)$
          $\omega \leftarrow (x_1, \frac{\eta}{\delta}(v_2 - u_2) + w_2)$ ▷ apply Equation (1)
        else if l.u.b.$(w, x) = x$ **then** ▷ Case $p_L(x) \geq p_L(w)$
          $\omega \leftarrow (x_1, \frac{\eta}{\delta}(u_2 - v_2) + w_2)$ ▷ apply Equation (6)
        else ▷ Case where l.u.b.$(w, x) \neq x$ or w
          $\omega \leftarrow (x_1, \frac{\eta}{\delta}(u_2 - v_2) + w_2)$ ▷ apply Equation (6)
          $\omega' \leftarrow (x_1, \frac{\eta}{\delta}(v_2 - u_2) + w_2)$ ▷ apply Equation (1)
          if CheckOmega(ω', u, v, w, x,Below) **then** ▷ Check if ω' satisfies condition 2 of Proposition 9
            Append(Ω, ω')
          end if
        end if
        if CheckOmega(ω, u, v, w, x,Below) **then** ▷ Checking if ω satisfies condition 2 of Proposition 9
          Append(Ω, ω)
        end if
      end if
      ▷ Case when x pushes right, u, v, w push up
      if CheckPts(CritPts,Above) **then** ▷ Checking if the configuration is possible with x above the line and u, v, w below the line
        if $u_1 == v_1$ **then continue** ▷ This matching has null cost.
        end if
        if $u_1 < v_1$ **then**
          Exchange u, v
        end if ▷ ensuring $p_L(u) > p_L(v)$
        if l.u.b.$(w, x) = w$ **then** ▷ Case where $p_L(w) \geq p_L(x)$
          $\omega \leftarrow (\frac{\eta}{\delta}(v_1 - u_1) + w_1, x_2)$ ▷ apply Equation (2)
        else if l.u.b.$(w, x) = x$ **then** ▷ Case where $p_L(x) \geq p_L(w)$
          $\omega \leftarrow (\frac{\eta}{\delta}(u_1 - v_1) + w_1, x_2)$ ▷ apply Equation (7)
        else ▷ Case where l.u.b.$(w, x) \neq x$ or w
          $\omega \leftarrow (\frac{\eta}{\delta}(v_1 - u_1) + w_1, x_2)$ ▷ apply Equation (2)
          $\omega' \leftarrow (\frac{\eta}{\delta}(u_1 - v_1) + w_1, x_2)$ ▷ apply Equation (7)
          if CheckOmega(ω', u, v, w, x,Above) **then** ▷ Checking if ω' satisfies condition 1 of Proposition 9
            Append(Ω, ω')
          end if
        end if
        if CheckOmega(ω, u, v, w, x,Above) **then** ▷ Checking if ω satisfies condition 1 of Proposition 9
          Append(Ω, ω)
        end if
      end if
    end for
  end for
  return Ω
end function

Algorithm 4 Function to assess if, for distinct four points x, u, v, w, there is a line L with positive slope separating x, w from u, v, where x, w push up, and u, v push right.

Input: u, v, w, x
Output: True if the configuration is good and False otherwise.

```
1: function CheckPts2(u, v, w, x)
2:
3:     Set Q and R as in Proposition 12
4:     if x ∈ Q or w ∈ Q then
5:         return False
6:     else if u ∈ R or v ∈ R then
7:         return False
8:     else
9:         return True
10:    end if
11: end function
```

Algorithm 5 (2paired-vs-2paired) An algorithm to compute switch points ω from a set of four points c_1, c_2, c_3, c_4 so that two paired points are one side of a with slope given by ω and the remaining two paired points are strictly on the other side.

Input: $c_1, c_2, c_3, c_4 \in C_M \sqcup C_N$. **Output:** The partial list of ω slopes generated by case 3 in the proof of [2, Lemma 4.1]
Ensure: there are 4 distinct points among $\{c_1, c_2, c_3, c_4\}$

```
1: function OmegaSlopes2v2paired(c1, c2, c3, c4)
2:     Ω ← ∅
3:     CritPts ← {c1, c2, c3, c4}
4:     for {c, d} in CritPts do                          ▷ Choose 2 of the 4 points
5:         x ← c
6:         w ← d
7:         if w1 < x1 then                               ▷ ensuring pL(w) > pL(x)
8:             Exchange w, x
9:         else if w1 == x1 then continue                ▷ This matching has null cost.
10:        end if
11:        u ← an element of CritPts\{x, w}
12:        v ← CritPts\{x, w, u}
13:        if u2 < v2 then                               ▷ ensuring pL(u) > pL(v)
14:            Exchange u, v
15:        else if u2 == v2 then continue                ▷ This matching has null cost.
16:        end if
17:        if x.parent() == w.parent() then              ▷ determining η
18:            η ← 2
19:        else
20:            η ← 1
21:        end if
22:        if u.parent() == v.parent() then              ▷ determining δ
23:            δ ← 2
24:        else
25:            δ ← 1
26:        end if
27:        if CheckPts2(x, w, u, v) then   ▷ Checking if the configuration is possible for labeled points
28:
29:            ω ← [0 : δ(w1 − x1) : η(u2 − v2)]          ▷ Apply Equation (3)
30:            m ← η(u2−v2) / δ(w1−x1)
31:            if max{x2 − mx1, w2 − mw1} < min{u2 − mu1, v2 − mv1} then ▷ Checking the condition
    of Proposition 13
32:                Append(Ω, ω)
33:            end if
34:        end if
35:    end for
36:    return Ω
37: end function
```

Algorithm 6 Functions to assess if ω satisfies the conditions of Proposition 14 and is not superfluous.

Input: ω, u, v, w, x. **Output:** True, if ω is correct and not superfluous, otherwise False.

function CheckOmega2(ω, u, v, w, x)

  for $p \in \{u, v, w, x\}$ **do** ▷ Assign ω quadrants and associated slopes and signs for relevant points.
    if $p_1 > \omega_1$ and $p_2 > \omega_2$ **then**
      $Q_p \leftarrow 1$
      $m_p \leftarrow \frac{p_2 - \omega_2}{p_1 - \omega_1}$
      if $p \in \{v, x\}$ **then**
        $\text{sign}_p \leftarrow 1$
      else
        $\text{sign}_p \leftarrow -1$
      end if
    else if $p_1 \leq \omega_1$ and $p_2 \geq \omega_2$ **then**
      $Q_p \leftarrow 2$
    else if $p_1 < \omega_1$ and $p_2 < \omega_2$ **then**
      $Q_p \leftarrow 3$
      $m_p \leftarrow \frac{p_2 - \omega_2}{p_1 - \omega_1}$
      if $p \in \{v, x\}$ **then**
        $\text{sign}_p \leftarrow -1$
      else
        $\text{sign}_p \leftarrow 1$
      end if
    else
      $Q_p \leftarrow 4$
    end if
  end for
  if $\omega \in \{x, v, u, w\}$ **then**
    return False
  else if $Q_x = 2$ or $Q_v = 2$ or $Q_u = 4$ or $Q_w = 4$ **then** ▷ Case 1 of Proposition 14
    return False
  else if $Q_x = 4$ and $Q_v = 4$ and $Q_u = 2$ and $Q_w = 2$ **then** ▷ Case 2 of Proposition 14
    return True
  else ▷ Case 3 of Proposition 14
    $P \leftarrow \emptyset$, $S \leftarrow \emptyset$
    for $p \in \{v, x\}$ **do**
      if $Q_p = 4$ **then continue** ▷ Determine if x or v are in $Q_1 \cup Q_3$
      else
        Append(P, p)
      end if
    end for
    for $p \in \{u, w\}$ **do**
      if $Q_p = 2$ **then continue** ▷ Determine if u or w are in $Q_1 \cup Q_3$
      else
        Append(S, p)
      end if
    end for
    for $p \in P \cup S$ **do** ▷ Determine slope intervals for all points in $P \cup S$
      if $\text{sign}_p = 1$ **then**
        $\text{int}_p \leftarrow (m_p, \infty)$
      else
        $\text{int}_p \leftarrow (0, m_p)$
      end if
    end for
    if $\cap_{p \in P} \text{int}_p \neq \emptyset$ **then** ▷ Determine existence of slope satisfying conditions of Proposition 14
      return True
    else
      return False
    end if
  end if
end function

Algorithm 7 (2unpaired-vs-2unpaired) An algorithm to compute switch points ω from a set of four points c_1, c_2, c_3, c_4 so that two unpaired points are one side of a with slope given by ω and the remaining two unpaired points are strictly on the other side.

Input: $c_1, c_2, c_3, c_4 \in C_M \sqcup C_N$. **Output:** The partial list of ω points generated by case 4 in the proof of [2, Lemma 4.1]

Ensure: there are 3 distinct points among $\{c_1, c_2, c_3, c_4\}$

function OmegaPts2v2unpaired(c_1, c_2, c_3, c_4)
  $\Omega \leftarrow \emptyset$
  CritPts $\leftarrow \{c_1, c_2, c_3, c_4\}$
  for $\{c, d\}$ in CritPts **do** ▷ Choose 2 of the 4 points
    $x \leftarrow c$
    $v \leftarrow d$
    for f in CritPts.remove(x, v) **do** ▷ take another point
      if $f == x$ **then continue**
      else
        $w \leftarrow f$
      end if
      for g in CritPts.remove(x, v, w) **do** ▷ take another point
        if $g == v$ **then continue**
        else
          $u \leftarrow g$
        end if
        if x.parent() == w.parent() **then** ▷ determining η
          $\eta \leftarrow 2$
        else
          $\eta \leftarrow 1$
        end if
        if u.parent() == v.parent() **then** ▷ determining δ
          $\delta \leftarrow 2$
        else
          $\delta \leftarrow 1$
        end if
        if CheckPts2(x, v, w, u) **then** ▷ Check if the configuration is possible
          if $\begin{cases} (\text{l.u.b.}(u,v) = u \text{ and l.u.b.}(x,w) = w) & \text{or} \\ (\text{l.u.b.}(u,v) = v \text{ and l.u.b.}(x,w) = x) & \text{or} \\ \text{l.u.b.}(u,v) \notin \{u,v\} & \text{or} \\ \text{l.u.b.}(x,w) \notin \{x,w\} \end{cases}$ **then**
            if $\delta = \eta$ and $v_1 \neq x_1$ and $u_2 \neq w_2$ **then**
              $\omega \leftarrow [0 : (v_1 - x_1) : (u_2 - w_2)]$ ▷ Apply Equation (4)
              $m \leftarrow \frac{u_2 - w_2}{v_1 - x_1}$ ▷ Check the condition of Proposition 13
              if $m > 0$ and $\max\{x_2 - mx_1, v_2 - mv_1\} < \min\{u_2 - mu_1, w_2 - mw_1\}$ **then**
                Append(Ω, ω)
              end if
            else if $\delta \neq \eta$ **then**
              $\omega \leftarrow \left(\frac{\eta v_1 - \delta x_1}{\eta - \delta}, \frac{\eta u_2 - \delta w_2}{\eta - \delta}\right)$ ▷ Apply Equation (5)
              if CheckOmega2(ω, u, v, w, x) **then** ▷ Check that some line through ω gives $\Delta_L = 0$
                Append(Ω, ω)
              end if
            end if
          end if
          if $\begin{cases} (\text{l.u.b.}(u,v) = u \text{ and l.u.b.}(x,w) = x) & \text{or} \\ (\text{l.u.b.}(u,v) = v \text{ and l.u.b.}(x,w) == w) & \text{or} \\ \text{l.u.b.}(u,v) \notin \{u,v\} & \text{or} \\ \text{l.u.b.}(x,w) \notin \{x,w\} \end{cases}$ **then**
            $\omega \leftarrow \left(\frac{\eta v_1 + \delta x_1}{\eta + \delta}, \frac{\eta u_2 + \delta w_2}{\eta + \delta}\right)$ ▷ Apply Equation (8)
            if CheckOmega2(ω, u, v, w, x) **then** ▷ Check that some line through ω gives $\Delta_L = 0$
              Append(Ω, ω)
            end if
          end if
        end if
      end for
    end for
  end for
  return Ω
end function